

Then $r\varphi \in \text{Hom}_R(M, N)$ and with this action of the commutative ring R the abelian group $\text{Hom}_R(M, N)$ is an R -module.

- (3) If $\varphi \in \text{Hom}_R(L, M)$ and $\psi \in \text{Hom}_R(M, N)$ then $\psi \circ \varphi \in \text{Hom}_R(L, N)$.
 (4) With addition as above and multiplication defined as function composition, $\text{Hom}_R(M, M)$ is a ring with 1. When R is commutative $\text{Hom}_R(M, M)$ is an R -algebra.

Proof: (1) Certainly $\varphi(rx+y) = r\varphi(x) + \varphi(y)$ if φ is an R -module homomorphism. Conversely, if $\varphi(rx+y) = r\varphi(x) + \varphi(y)$, take $r = 1$ to see that φ is additive and take $y = 0$ to see that φ commutes with the action of R on M (i.e., is *homogeneous*).

(2) It is straightforward to check that all the abelian group and R -module axioms hold with these definitions — the details are left as an exercise. We note that the commutativity of R is used to show that $r\varphi$ satisfies the second axiom of an R -module homomorphism, namely,

$$\begin{aligned}(r_1\varphi)(r_2m) &= r_1\varphi(r_2m) && \text{(by definition of } r_1\varphi) \\ &= r_1r_2(\varphi(m)) && \text{(since } \varphi \text{ is a homomorphism)} \\ &= r_2r_1\varphi(m) && \text{(since } R \text{ is commutative)} \\ &= r_2(r_1\varphi)(m) && \text{(by definition of } r_1\varphi).\end{aligned}$$

Verification of the axioms relies ultimately on the hypothesis that N is an R -module. The domain M could in fact be any set — it does not have to be an R -module nor an abelian group.

- (3) Let φ and ψ be as given and let $r \in R$, $x, y \in L$. Then

$$\begin{aligned}(\psi \circ \varphi)(rx+y) &= \psi(\varphi(rx+y)) \\ &= \psi(r\varphi(x) + \varphi(y)) && \text{(by (1) applied to } \varphi) \\ &= r\psi(\varphi(x)) + \psi(\varphi(y)) && \text{(by (1) applied to } \psi) \\ &= r(\psi \circ \varphi)(x) + (\psi \circ \varphi)(y)\end{aligned}$$

so, by (1), $\psi \circ \varphi$ is an R -module homomorphism.

(4) Note that since the domain and codomain of the elements of $\text{Hom}_R(M, M)$ are the same, function composition is defined. By (3), it is a binary operation on $\text{Hom}_R(M, M)$. As usual, function composition is associative. The remaining ring axioms are straightforward to check — the details are left as an exercise. The identity function, I , (as usual, $I(x) = x$, for all $x \in M$) is seen to be the multiplicative identity of $\text{Hom}_R(M, M)$. If R is commutative, then (2) shows that the ring $\text{Hom}_R(M, M)$ is a left R -module and defining $\varphi r = r\varphi$ for all $\varphi \in \text{Hom}_R(M, M)$ and $r \in R$ makes $\text{Hom}_R(M, M)$ into an R -algebra.

Definition. The ring $\text{Hom}_R(M, M)$ is called the *endomorphism ring of M* and will often be denoted by $\text{End}_R(M)$, or just $\text{End}(M)$ when the ring R is clear from the context. Elements of $\text{End}(M)$ are called *endomorphisms*.

When R is commutative there is a natural map from R into $\text{End}(M)$ given by $r \mapsto rI$, where the latter endomorphism of M is just multiplication by r on M (cf. Exercise 7). The image of R is contained in the center of $\text{End}(M)$ so if R has an identity, $\text{End}(M)$ is an R -algebra. The ring homomorphism (cf. Exercise 7) from R to $\text{End}_R(M)$ may not be injective since for some r we may have $rm = 0$ for all $m \in M$ (e.g., $R = \mathbb{Z}$, $M = \mathbb{Z}/2\mathbb{Z}$, and $r = 2$). When R is a field, however, this map is injective (in general, no unit is in the kernel of this map) and the copy of R in $\text{End}_R(M)$ is called the (subring of) *scalar transformations*.

Next we prove that every submodule N of an R -module M is “normal” in the sense that we can *always* form the quotient module M/N , and the natural projection $\pi : M \rightarrow M/N$ is an R -module homomorphism with kernel N . The proof of this fact and, more generally, the subsequent proofs of the isomorphism theorems for modules follow easily from the corresponding facts for groups. The reason for this is because a module is first of all an *abelian* group and so *every* submodule is automatically a normal subgroup and any module homomorphism is, in particular, a homomorphism of abelian groups, all of which we have already considered in Chapter 3. What remains to be proved in order to extend results on abelian groups to corresponding results on modules is to check that the action of R is compatible with these group quotients and homomorphisms. For example, the map π above was shown to be a group homomorphism in Chapter 3 but the abelian group M/N must be shown to be an R -module (i.e., to have an action by R) and property (b) in the definition of a module homomorphism must be checked for π .

Proposition 3. Let R be a ring, let M be an R -module and let N be a submodule of M . The (additive, abelian) quotient group M/N can be made into an R -module by defining an action of elements of R by

$$r(x + N) = (rx) + N, \quad \text{for all } r \in R, x + N \in M/N.$$

The natural projection map $\pi : M \rightarrow M/N$ defined by $\pi(x) = x + N$ is an R -module homomorphism with kernel N .

Proof: Since M is an abelian group under $+$ the quotient group M/N is defined and is an abelian group. To see that the action of the ring element r on the coset $x + N$ is well defined, suppose $x + N = y + N$, i.e., $x - y \in N$. Since N is a (left) R -submodule, $r(x - y) \in N$. Thus $rx - ry \in N$ and $rx + N = ry + N$, as desired. Now since the operations in M/N are “compatible” with those of M , the axioms for an R -module are easily checked in the same way as was done for quotient groups. For example, axiom 2(b) holds as follows: for all $r_1, r_2 \in R$ and $x + N \in M/N$, by definition of the action of ring elements on elements of M/N

$$\begin{aligned} (r_1 r_2)(x + N) &= (r_1 r_2 x) + N \\ &= r_1(r_2 x + N) \\ &= r_1(r_2(x + N)). \end{aligned}$$

The other axioms are similarly checked — the details are left as an exercise. Finally, the natural projection map π described above is, in particular, the natural projection of the abelian group M onto the abelian group M/N hence is a group homomorphism with kernel N . The kernel of any module homomorphism is the same as its kernel when viewed as a homomorphism of the abelian group structures. It remains only to show π is a module homomorphism, i.e., $\pi(rm) = r\pi(m)$. But

$$\begin{aligned}\pi(rm) &= rm + N \\ &= r(m + N) \quad (\text{by definition of the action of } R \text{ on } M/N) \\ &= r\pi(m).\end{aligned}$$

This completes the proof.

All the isomorphism theorems stated for groups also hold for R -modules. The proofs are similar to that of Proposition 3 above in that they begin by invoking the corresponding theorem for groups and then prove that the group homomorphisms are also R -module homomorphisms. To state the Second Isomorphism Theorem we need the following.

Definition. Let A, B be submodules of the R -module M . The *sum* of A and B is the set

$$A + B = \{a + b \mid a \in A, b \in B\}.$$

One can easily check that the sum of two submodules A and B is a submodule and is the smallest submodule which contains both A and B .

Theorem 4. (Isomorphism Theorems)

- (1) *(The First Isomorphism Theorem for Modules)* Let M, N be R -modules and let $\varphi : M \rightarrow N$ be an R -module homomorphism. Then $\ker \varphi$ is a submodule of M and $M / \ker \varphi \cong \varphi(M)$.
- (2) *(The Second Isomorphism Theorem)* Let A, B be submodules of the R -module M . Then $(A + B)/B \cong A/(A \cap B)$.
- (3) *(The Third Isomorphism Theorem)* Let M be an R -module, and let A and B be submodules of M with $A \subseteq B$. Then $(M/A)/(B/A) \cong M/B$.
- (4) *(The Fourth or Lattice Isomorphism Theorem)* Let N be a submodule of the R -module M . There is a bijection between the submodules of M which contain N and the submodules of M/N . The correspondence is given by $A \leftrightarrow A/N$, for all $A \supseteq N$. This correspondence commutes with the processes of taking sums and intersections (i.e., is a lattice isomorphism between the lattice of submodules of M/N and the lattice of submodules of M which contain N).

Proof: Exercise.

EXERCISES

In these exercises R is a ring with 1 and M is a left R -module.

1. Use the submodule criterion to show that kernels and images of R -module homomorphisms are submodules.
2. Show that the relation “is R -module isomorphic to” is an equivalence relation on any set of R -modules.
3. Give an explicit example of a map from one R -module to another which is a group homomorphism but not an R -module homomorphism.
4. Let A be any $\mathbb{Z}$ -module, let a be any element of A and let n be a positive integer. Prove that the map $\varphi_a : \mathbb{Z}/n\mathbb{Z} \rightarrow A$ given by $\varphi(\bar{k}) = ka$ is a well defined $\mathbb{Z}$ -module homomorphism if and only if $na = 0$. Prove that $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, A) \cong A_n$, where $A_n = \{a \in A \mid na = 0\}$ (so A_n is the annihilator in A of the ideal (n) of $\mathbb{Z}$ — cf. Exercise 10, Section 1).
5. Exhibit all $\mathbb{Z}$ -module homomorphisms from $\mathbb{Z}/30\mathbb{Z}$ to $\mathbb{Z}/21\mathbb{Z}$.
6. Prove that $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/(n, m)\mathbb{Z}$.
7. Let z be a fixed element of the center of R . Prove that the map $m \mapsto zm$ is an R -module homomorphism from M to itself. Show that for a commutative ring R the map from R to $\text{End}_R(M)$ given by $r \mapsto rI$ is a ring homomorphism (where I is the identity endomorphism).
8. Let $\varphi : M \rightarrow N$ be an R -module homomorphism. Prove that $\varphi(\text{Tor}(M)) \subseteq \text{Tor}(N)$ (cf. Exercise 8 in Section 1).
9. Let R be a commutative ring. Prove that $\text{Hom}_R(R, M)$ and M are isomorphic as left R -modules. [Show that each element of $\text{Hom}_R(R, M)$ is determined by its value on the identity of R .]
10. Let R be a commutative ring. Prove that $\text{Hom}_R(R, R)$ and R are isomorphic as rings.
11. Let $A_1, A_2, \dots, A_n$ be R -modules and let B_i be a submodule of A_i for each $i = 1, 2, \dots, n$. Prove that

$$(A_1 \times \cdots \times A_n)/(B_1 \times \cdots \times B_n) \cong (A_1/B_1) \times \cdots \times (A_n/B_n).$$

[Recall Exercise 14 in Section 5.1.]

12. Let I be a left ideal of R and let n be a positive integer. Prove

$$R^n/IR^n \cong R/IR \times \cdots \times R/IR \quad (n \text{ times})$$

where IR^n is defined as in Exercise 5 of Section 1. [Use the preceding exercise.]

13. Let I be a nilpotent ideal in a commutative ring R (cf. Exercise 37, Section 7.3), let M and N be R -modules and let $\varphi : M \rightarrow N$ be an R -module homomorphism. Show that if the induced map $\bar{\varphi} : M/IM \rightarrow N/IN$ is surjective, then φ is surjective.
14. Let $R = \mathbb{Z}[x]$ be the ring of polynomials in x and let $A = \mathbb{Z}[t_1, t_2, \dots]$ be the ring of polynomials in the independent indeterminates $t_1, t_2, \dots$. Define an action of R on A as follows: 1) let $1 \in R$ act on A as the identity, 2) for $n \geq 1$ let $x^n \circ 1 = t_n$, let $x^n \circ t_i = t_{n+i}$ for $i = 1, 2, \dots$, and let x^n act as 0 on monomials in A of (total) degree at least two, and 3) extend $\mathbb{Z}$ -linearly, i.e., so that the module axioms 2(a) and 2(c) are satisfied.
 - (a) Show that $x^{p+q} \circ t_i = x^p \circ (x^q \circ t_i) = t_{p+q+i}$ and use this to show that under this action the ring A is a (unital) R -module.
 - (b) Show that the map $\varphi : R \rightarrow A$ defined by $\varphi(r) = r \circ 1_A$ is an R -module homomorphism of the ring R into the ring A mapping 1_R to 1_A , but is not a ring homomorphism from R to A .

10.3 GENERATION OF MODULES, DIRECT SUMS, AND FREE MODULES

Let R be a ring with 1. As in the preceding sections the term “module” will mean “left module.” We first extend the notion of the sum of two submodules to sums of any finite number of submodules and define the submodule generated by a subset.

Definition. Let M be an R -module and let $N_1, \dots, N_n$ be submodules of M .

- (1) The *sum* of $N_1, \dots, N_n$ is the set of all finite sums of elements from the sets N_i : $\{a_1 + a_2 + \dots + a_n \mid a_i \in N_i \text{ for all } i\}$. Denote this sum by $N_1 + \dots + N_n$.
- (2) For any subset A of M let

$$RA = \{r_1a_1 + r_2a_2 + \dots + r_ma_m \mid r_1, \dots, r_m \in R, a_1, \dots, a_m \in A, m \in \mathbb{Z}^+\}$$

(where by convention $RA = \{0\}$ if $A = \emptyset$). If A is the finite set $\{a_1, a_2, \dots, a_n\}$ we shall write $Ra_1 + Ra_2 + \dots + Ra_n$ for RA . Call RA the *submodule of M generated by A* . If N is a submodule of M (possibly $N = M$) and $N = RA$, for some subset A of M , we call A a *set of generators* or *generating set* for N , and we say N is *generated by A* .

- (3) A submodule N of M (possibly $N = M$) is *finitely generated* if there is some finite subset A of M such that $N = RA$, that is, if N is generated by some finite subset.
- (4) A submodule N of M (possibly $N = M$) is *cyclic* if there exists an element $a \in M$ such that $N = Ra$, that is, if N is generated by one element:

$$N = Ra = \{ra \mid r \in R\}.$$

Note that these definitions do not require that the ring R contain a 1, however this condition ensures that A is contained in RA . It is easy to see using the Submodule Criterion that for any subset A of M , RA is indeed a submodule of M and is the smallest submodule of M which contains A (i.e., any submodule of M which contains A also contains RA). In particular, for submodules $N_1, \dots, N_n$ of M , $N_1 + \dots + N_n$ is just the submodule generated by the set $N_1 \cup \dots \cup N_n$ and is the smallest submodule of M containing N_i , for all i . If $N_1, \dots, N_n$ are generated by sets $A_1, \dots, A_n$ respectively, then $N_1 + \dots + N_n$ is generated by $A_1 \cup \dots \cup A_n$. Note that cyclic modules are, a fortiori, finitely generated.

A submodule N of an R -module M may have many different generating sets (for instance the set N itself always generates N). If N is finitely generated, then there is a smallest nonnegative integer d such that N is generated by d elements (and no fewer). Any generating set consisting of d elements will be called a *minimal set of generators for N* (it is not unique in general). If N is not finitely generated, it need not have a minimal generating set.

The process of generating submodules of an R -module M by taking subsets A of M and forming all finite “ R -linear combinations” of elements of A will be our primary way of producing submodules (this notion is perhaps familiar from vector space theory where it is referred to as taking the *span* of A). The obstruction which made the analogous process so difficult for groups in general was the noncommutativity of group

operations. For abelian groups, G , however, it was much simpler to control the subgroup $\langle A \rangle$ generated by A , for a subset A of G (see Section 2.4 for the complete discussion of this). The situation for R -modules is similar to that of abelian groups (even if R is a noncommutative ring) because we can always collect “like terms” in elements of A , i.e., terms such as $r_1a_1 + r_2a_2 + s_1a_1$ can always be simplified to $(r_1 + s_1)a_1 + r_2a_2$. This again reflects the underlying abelian group structure of modules.

Examples

- (1) Let $R = \mathbb{Z}$ and let M be any R -module, that is, any abelian group. If $a \in M$, then $\mathbb{Z}a$ is just the cyclic subgroup of M generated by a : $\langle a \rangle$ (compare Definition 4 above with the definition of a cyclic group). More generally, M is generated as a $\mathbb{Z}$ -module by a set A if and only if M is generated as a group by A (that is, the action of ring elements in this instance produces no elements that cannot already be obtained from A by addition and subtraction). The definition of finitely generated for $\mathbb{Z}$ -modules is identical to that for abelian groups found in Chapter 5.
- (2) Let R be a ring with 1 and let M be the (left) R -module R itself. Note that R is a finitely generated, in fact cyclic, R -module because $R = R1$ (i.e., we can take $A = \{1\}$). Recall that the submodules of R are precisely the left ideals of R , so saying I is a cyclic R -submodule of the left R -module R is the same as saying I is a principal ideal of R (usually the term “principal ideal” is used in the context of commutative rings). Also, saying I is a finitely generated R -submodule of R is the same as saying I is a finitely generated ideal. When R is a commutative ring we often write AR or aR for the submodule (ideal) generated by A or a respectively, as we have been doing for $\mathbb{Z}$ when we wrote $n\mathbb{Z}$. In this situation $AR = RA$ and $aR = Ra$ (elementwise as well). Thus a Principal Ideal Domain is a (commutative) integral domain R with identity in which every R -submodule of R is cyclic.

Submodules of a finitely generated module need not be finitely generated: take M to be the cyclic R -module R itself where R is the polynomial ring in infinitely many variables $x_1, x_2, x_3, \dots$ with coefficients in some field F . The submodule (i.e., 2-sided ideal) generated by $\{x_1, x_2, \dots\}$ cannot be generated by any finite set (note that one must show that *no* finite subset of this ideal will generate it).

- (3) Let R be a ring with 1 and let M be the free module of rank n over R , as described in the first section. For each $i \in \{1, 2, \dots, n\}$ let $e_i = (0, 0, \dots, 0, 1, 0, \dots, 0)$, where the 1 appears in position i . Since

$$(s_1, s_2, \dots, s_n) = \sum_{i=1}^n s_i e_i$$

it is clear that M is generated by $\{e_1, \dots, e_n\}$. If R is commutative then this is a *minimal* generating set (cf. Exercises 2 and 27).

- (4) Let F be a field, let x be an indeterminate, let V be a vector space over F and let T be a linear transformation from V to V . Make V into an $F[x]$ -module via T . Then V is a *cyclic* $F[x]$ -module (with generator v) if and only if $V = \{p(x)v \mid p(x) \in F[x]\}$, that is, if and only if every element of V can be written as an F -linear combination of elements of the set $\{T^n(v) \mid n \geq 0\}$. This in turn is equivalent to saying $\{v, T(v), T^2(v), \dots\}$ span V as a vector space over F .

For instance if T is the identity linear transformation from V to V or the zero linear transformation, then for every $v \in V$ and every $p(x) \in F[x]$ we have $p(x)v = \alpha v$ for some $\alpha \in F$. Thus if V has dimension > 1 , V cannot be a cyclic $F[x]$ -module.

For another example suppose V is affine n -space and T is the “shift operator” described in Section 1. Let e_i be the i^{th} basis vector (as usual) numbered so that T is defined by $T^k(e_n) = e_{n-k}$ for $1 \leq k < n$. Thus V is spanned by the elements $e_n, T(e_n), \dots, T^{n-1}(e_n)$, that is, V is a cyclic $F[x]$ -module with generator e_n . For $n > 1$, V is not, however, a cyclic F -module (i.e., is not a 1-dimensional vector space over F).

Definition. Let $M_1, \dots, M_k$ be a collection of R -modules. The collection of k -tuples $(m_1, m_2, \dots, m_k)$ where $m_i \in M_i$ with addition and action of R defined componentwise is called the *direct product* of $M_1, \dots, M_k$, denoted $M_1 \times \dots \times M_k$.

It is evident that the direct product of a collection of R -modules is again an R -module. The direct product of $M_1, \dots, M_k$ is also referred to as the (*external*) *direct sum* of $M_1, \dots, M_k$ and denoted $M_1 \oplus \dots \oplus M_k$. The direct product and direct sum of an infinite number of modules (which are different in general) are defined in Exercise 20.

The next proposition indicates when a module is isomorphic to the direct product of some of its submodules and is the analogue for modules of Theorem 9 in Section 5.4 (which determines when a group is the direct product of two of its subgroups).

Proposition 5. Let $N_1, N_2, \dots, N_k$ be submodules of the R -module M . Then the following are equivalent:

(1) The map $\pi : N_1 \times N_2 \times \dots \times N_k \rightarrow N_1 + N_2 + \dots + N_k$ defined by

$$\pi(a_1, a_2, \dots, a_k) = a_1 + a_2 + \dots + a_k$$

is an isomorphism (of R -modules): $N_1 + N_2 + \dots + N_k \cong N_1 \times N_2 \times \dots \times N_k$.

(2) $N_j \cap (N_1 + N_2 + \dots + N_{j-1} + N_{j+1} + \dots + N_k) = 0$ for all $j \in \{1, 2, \dots, k\}$.

(3) Every $x \in N_1 + \dots + N_k$ can be written *uniquely* in the form $a_1 + a_2 + \dots + a_k$ with $a_i \in N_i$.

Proof: To prove (1) implies (2), suppose for some j that (2) fails to hold and let $a_j \in (N_1 + \dots + N_{j-1} + N_{j+1} + \dots + N_k) \cap N_j$, with $a_j \neq 0$. Then

$$a_j = a_1 + \dots + a_{j-1} + a_{j+1} + \dots + a_k$$

for some $a_i \in N_i$, and $(a_1, \dots, a_{j-1}, -a_j, a_{j+1}, \dots, a_k)$ would be a nonzero element of $\ker \pi$, a contradiction.

Assume now that (2) holds. If for some module elements $a_i, b_i \in N_i$ we have

$$a_1 + a_2 + \dots + a_k = b_1 + b_2 + \dots + b_k$$

then for each j we have

$$a_j - b_j = (b_1 - a_1) + \dots + (b_{j-1} - a_{j-1}) + (b_{j+1} - a_{j+1}) + \dots + (b_k - a_k).$$

The left hand side is in N_j and the right side belongs to $N_1 + \dots + N_{j-1} + N_{j+1} + \dots + N_k$. Thus

$$a_j - b_j \in N_j \cap (N_1 + \dots + N_{j-1} + N_{j+1} + \dots + N_k) = 0.$$

This shows $a_j = b_j$ for all j , and so (2) implies (3).

Finally, to see that (3) implies (1) observe first that the map π is clearly a surjective R -module homomorphism. Then (3) simply implies π is injective, hence is an isomorphism, completing the proof.

If an R -module $M = N_1 + N_2 + \cdots + N_k$ is the sum of submodules $N_1, N_2, \dots, N_k$ of M satisfying the equivalent conditions of the proposition above, then M is said to be the (internal) direct sum of $N_1, N_2, \dots, N_k$, written

$$M = N_1 \oplus N_2 \oplus \cdots \oplus N_k.$$

By the proposition, this is equivalent to the assertion that every element m of M can be written *uniquely* as a sum of elements $m = n_1 + n_2 + \cdots + n_k$ with $n_i \in N_i$. (Note that part (1) of the proposition is the statement that the internal direct sum of $N_1, N_2, \dots, N_k$ is isomorphic to their external direct sum, which is the reason we identify them and use the same notation for both.)

Definition. An R -module F is said to be *free* on the subset A of F if for every nonzero element x of F , there exist unique nonzero elements $r_1, r_2, \dots, r_n$ of R and unique $a_1, a_2, \dots, a_n$ in A such that $x = r_1 a_1 + r_2 a_2 + \cdots + r_n a_n$, for some $n \in \mathbb{Z}^+$. In this situation we say A is a *basis* or *set of free generators* for F . If R is a commutative ring the cardinality of A is called the *rank* of F (cf. Exercise 27).

One should be careful to note the difference between the uniqueness property of direct sums (Proposition 5(3)) and the uniqueness property of free modules. Namely, in the direct sum of two modules, say $N_1 \oplus N_2$, each element can be written uniquely as $n_1 + n_2$; here the uniqueness refers to the *module elements* n_1 and n_2 . In the case of free modules, the uniqueness is on the *ring elements as well as the module elements*. For example, if $R = \mathbb{Z}$ and $N_1 = N_2 = \mathbb{Z}/2\mathbb{Z}$, then each element of $N_1 \oplus N_2$ has a unique representation in the form $n_1 + n_2$ where each $n_i \in N_i$, however n_1 (for instance) can be expressed as n_1 or $3n_1$ or $5n_1 \dots$ etc., so each element does not have a unique representation in the form $r_1 a_1 + r_2 a_2$, where $r_1, r_2 \in R, a_1 \in N_1$ and $a_2 \in N_2$. Thus $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ is not a free $\mathbb{Z}$ -module on the set $\{(1, 0), (0, 1)\}$. Similarly, it is not free on any set.

Theorem 6. For any set A there is a free R -module $F(A)$ on the set A and $F(A)$ satisfies the following *universal property*: if M is any R -module and $\varphi : A \rightarrow M$ is any map of sets, then there is a unique R -module homomorphism $\Phi : F(A) \rightarrow M$ such that $\Phi(a) = \varphi(a)$, for all $a \in A$, that is, the following diagram commutes.

$$\begin{array}{ccc} A & \xrightarrow{\text{inclusion}} & F(A) \\ & \searrow \varphi & \downarrow \Phi \\ & & M \end{array}$$

When A is the finite set $\{a_1, a_2, \dots, a_n\}$, $F(A) = Ra_1 \oplus Ra_2 \oplus \cdots \oplus Ra_n \cong R^n$. (Compare: Section 6.3, free groups.)

Proof: Let $F(A) = \{0\}$ if $A = \emptyset$. If A is nonempty let $F(A)$ be the collection of all set functions $f : A \rightarrow R$ such that $f(a) = 0$ for all but finitely many $a \in A$. Make

$F(A)$ into an R -module by pointwise addition of functions and pointwise multiplication of a ring element times a function, i.e.,

$$(f + g)(a) = f(a) + g(a) \quad \text{and} \\ (rf)(a) = r(f(a)), \quad \text{for all } a \in A, r \in R \text{ and } f, g \in F(A).$$

It is an easy matter to check that all the R -module axioms hold (the details are omitted). Identify A as a subset of $F(A)$ by $a \mapsto f_a$, where f_a is the function which is 1 at a and zero elsewhere. We can, in this way, think of $F(A)$ as all finite R -linear combinations of elements of A by identifying each function f with the sum $r_1a_1 + r_2a_2 + \cdots + r_na_n$, where f takes on the value r_i at a_i and is zero at all other elements of A . Moreover, each element of $F(A)$ has a unique expression as such a formal sum. To establish the universal property of $F(A)$ suppose $\varphi : A \rightarrow M$ is a map of the set A into the R -module M . Define $\Phi : F(A) \rightarrow M$ by

$$\Phi : \sum_{i=1}^n r_i a_i \mapsto \sum_{i=1}^n r_i \varphi(a_i).$$

By the uniqueness of the expression for the elements of $F(A)$ as linear combinations of the a_i we see easily that Φ is a well defined R -module homomorphism (the details are left as an exercise). By definition, the restriction of Φ to A equals φ . Finally, since $F(A)$ is generated by A , once we know the values of an R -module homomorphism on A its values on every element of $F(A)$ are uniquely determined, so Φ is the unique extension of φ to all of $F(A)$.

When A is the finite set $\{a_1, a_2, \dots, a_n\}$ Proposition 5(3) shows that $F(A) = Ra_1 \oplus Ra_2 \oplus \cdots \oplus Ra_n$. Since $R \cong Ra_i$ for all i (under the map $r \mapsto ra_i$) Proposition 5(1) shows that the direct sum is isomorphic to R^n .

Corollary 7.

- (1) If F_1 and F_2 are free modules on the same set A , there is a unique isomorphism between F_1 and F_2 which is the identity map on A .
- (2) If F is any free R -module with basis A , then $F \cong F(A)$. In particular, F enjoys the same universal property with respect to A as $F(A)$ does in Theorem 6.

Proof: Exercise.

If F is a free R -module with basis A , we shall often (particularly in the case of vector spaces) define R -module homomorphisms from F into other R -modules simply by specifying their values on the elements of A and then saying “*extend by linearity*.” Corollary 7(2) ensures that this is permissible.

When $R = \mathbb{Z}$, the free module on a set A is called the *free abelian group on A* . If $|A| = n$, $F(A)$ is called the free abelian group of *rank n* and is isomorphic to $\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}$ (n times). These definitions agree with the ones given in Chapter 5.

EXERCISES

In these exercises R is a ring with 1 and M is a left R -module.

1. Prove that if A and B are sets of the same cardinality, then the free modules $F(A)$ and $F(B)$ are isomorphic.
2. Assume R is commutative. Prove that $R^n \cong R^m$ if and only if $n = m$, i.e., two free R -modules of finite rank are isomorphic if and only if they have the same rank. [Apply Exercise 12 of Section 2 with I a maximal ideal of R . You may assume that if F is a field, then $F^n \cong F^m$ if and only if $n = m$, i.e., two finite dimensional vector spaces over F are isomorphic if and only if they have the same dimension — this will be proved later in Section 11.1.]
3. Show that the $F[x]$ -modules in Exercises 18 and 19 of Section 1 are both cyclic.
4. An R -module M is called a *torsion* module if for each $m \in M$ there is a nonzero element $r \in R$ such that $rm = 0$, where r may depend on m (i.e., $M = \text{Tor}(M)$ in the notation of Exercise 8 of Section 1). Prove that every finite abelian group is a torsion $\mathbb{Z}$ -module. Give an example of an infinite abelian group that is a torsion $\mathbb{Z}$ -module.
5. Let R be an integral domain. Prove that every finitely generated torsion R -module has a nonzero annihilator i.e., there is a nonzero element $r \in R$ such that $rm = 0$ for all $m \in M$ — here r does not depend on m (the annihilator of a module was defined in Exercise 9 of Section 1). Give an example of a torsion R -module whose annihilator is the zero ideal.
6. Prove that if M is a finitely generated R -module that is generated by n elements then every quotient of M may be generated by n (or fewer) elements. Deduce that quotients of cyclic modules are cyclic.
7. Let N be a submodule of M . Prove that if both M/N and N are finitely generated then so is M .
8. Let S be the collection of sequences $(a_1, a_2, a_3, \dots)$ of integers $a_1, a_2, a_3, \dots$ where all but finitely many of the a_i are 0 (called the *direct sum* of infinitely many copies of $\mathbb{Z}$). Recall that S is a ring under componentwise addition and multiplication and S does not have a multiplicative identity — cf. Exercise 20, Section 7.1. Prove that S is not finitely generated as a module over itself.
9. An R -module M is called *irreducible* if $M \neq 0$ and if 0 and M are the only submodules of M . Show that M is irreducible if and only if $M \neq 0$ and M is a cyclic module with any nonzero element as generator. Determine all the irreducible $\mathbb{Z}$ -modules.
10. Assume R is commutative. Show that an R -module M is irreducible if and only if M is isomorphic (as an R -module) to R/I where I is a maximal ideal of R . [By the previous exercise, if M is irreducible there is a natural map $R \rightarrow M$ defined by $r \mapsto rm$, where m is any fixed nonzero element of M .]
11. Show that if M_1 and M_2 are irreducible R -modules, then any nonzero R -module homomorphism from M_1 to M_2 is an isomorphism. Deduce that if M is irreducible then $\text{End}_R(M)$ is a division ring (this result is called *Schur's Lemma*). [Consider the kernel and the image.]
12. Let R be a commutative ring and let A, B and M be R -modules. Prove the following isomorphisms of R -modules:
 - (a) $\text{Hom}_R(A \times B, M) \cong \text{Hom}_R(A, M) \times \text{Hom}_R(B, M)$
 - (b) $\text{Hom}_R(M, A \times B) \cong \text{Hom}_R(M, A) \times \text{Hom}_R(M, B)$.
13. Let R be a commutative ring and let F be a free R -module of finite rank. Prove the following isomorphism of R -modules: $\text{Hom}_R(F, R) \cong F$.

14. Let R be a commutative ring and let F be the free R -module of rank n . Prove that $\text{Hom}_R(F, M) \cong M \times \cdots \times M$ (n times). [Use Exercise 9 in Section 2 and Exercise 12.]
15. An element $e \in R$ is called a *central idempotent* if $e^2 = e$ and $er = re$ for all $r \in R$. If e is a central idempotent in R , prove that $M = eM \oplus (1-e)M$. [Recall Exercise 14 in Section 1.]

The next two exercises establish the Chinese Remainder Theorem for modules (cf. Section 7.6).

16. For any ideal I of R let IM be the submodule defined in Exercise 5 of Section 1. Let $A_1, \dots, A_k$ be any ideals in the ring R . Prove that the map

$$M \rightarrow M/A_1M \times \cdots \times M/A_kM \quad \text{defined by} \quad m \mapsto (m + A_1M, \dots, m + A_kM)$$

is an R -module homomorphism with kernel $A_1M \cap A_2M \cap \cdots \cap A_kM$.

17. In the notation of the preceding exercise, assume further that the ideals $A_1, \dots, A_k$ are pairwise comaximal (i.e., $A_i + A_j = R$ for all $i \neq j$). Prove that

$$M/(A_1 \cdots A_k)M \cong M/A_1M \times \cdots \times M/A_kM.$$

[See the proof of the Chinese Remainder Theorem for rings in Section 7.6.]

18. Let R be a Principal Ideal Domain and let M be an R -module that is annihilated by the nonzero, proper ideal (a) . Let $a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ be the unique factorization of a into distinct prime powers in R . Let M_i be the annihilator of $p_i^{\alpha_i}$ in M , i.e., M_i is the set $\{m \in M \mid p_i^{\alpha_i} m = 0\}$ — called the *p_i -primary component* of M . Prove that

$$M = M_1 \oplus M_2 \oplus \cdots \oplus M_k.$$

19. Show that if M is a finite abelian group of order $a = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ then, considered as a $\mathbb{Z}$ -module, M is annihilated by (a) , the p_i -primary component of M is the unique Sylow p_i -subgroup of M and M is isomorphic to the direct product of its Sylow subgroups.
20. Let I be a nonempty index set and for each $i \in I$ let M_i be an R -module. The *direct product* of the modules M_i is defined to be their direct product as abelian groups (cf. Exercise 15 in Section 5.1) with the action of R componentwise multiplication. The *direct sum* of the modules M_i is defined to be the restricted direct product of the abelian groups M_i (cf. Exercise 17 in Section 5.1) with the action of R componentwise multiplication. In other words, the direct sum of the M_i 's is the subset of the direct product, $\prod_{i \in I} M_i$, which consists of all elements $\prod_{i \in I} m_i$ such that only finitely many of the components m_i are nonzero; the action of R on the direct product or direct sum is given by $r \prod_{i \in I} m_i = \prod_{i \in I} r m_i$ (cf. Appendix I for the definition of Cartesian products of infinitely many sets). The direct sum will be denoted by $\bigoplus_{i \in I} M_i$.
- (a) Prove that the direct product of the M_i 's is an R -module and the direct sum of the M_i 's is a submodule of their direct product.
- (b) Show that if $R = \mathbb{Z}$, $I = \mathbb{Z}^+$ and M_i is the cyclic group of order i for each i , then the direct sum of the M_i 's is not isomorphic to their direct product. [Look at torsion.]
21. Let I be a nonempty index set and for each $i \in I$ let N_i be a submodule of M . Prove that the following are equivalent:
- (i) the submodule of M generated by all the N_i 's is isomorphic to the direct sum of the N_i 's
 - (ii) if $\{i_1, i_2, \dots, i_k\}$ is any finite subset of I then $N_{i_1} \cap (N_{i_2} + \cdots + N_{i_k}) = 0$
 - (iii) if $\{i_1, i_2, \dots, i_k\}$ is any finite subset of I then $N_1 + \cdots + N_k = N_1 \oplus \cdots \oplus N_k$
 - (iv) for every element x of the submodule of M generated by the N_i 's there are unique elements $a_i \in N_i$ for all $i \in I$ such that all but a finite number of the a_i are zero and x is the (finite) sum of the a_i .

22. Let R be a Principal Ideal Domain, let M be a torsion R -module (cf. Exercise 4) and let p be a prime in R (do not assume M is finitely generated, hence it need not have a nonzero annihilator — cf. Exercise 5). The p -primary component of M is the set of all elements of M that are annihilated by some positive power of p .
- (a) Prove that the p -primary component is a submodule. [See Exercise 13 in Section 1.]
 - (b) Prove that this definition of p -primary component agrees with the one given in Exercise 18 when M has a nonzero annihilator.
 - (c) Prove that M is the (possibly infinite) direct sum of its p -primary components, as p runs over all primes of R .
23. Show that any direct sum of free R -modules is free.
24. (*An arbitrary direct product of free modules need not be free*) For each positive integer i let M_i be the free $\mathbb{Z}$ -module $\mathbb{Z}$, and let M be the direct product $\prod_{i \in \mathbb{Z}^+} M_i$ (cf. Exercise 20). Each element of M can be written uniquely in the form $(a_1, a_2, a_3, \dots)$ with $a_i \in \mathbb{Z}$ for all i . Let N be the submodule of M consisting of all such tuples with only finitely many nonzero a_i . Assume M is a free $\mathbb{Z}$ -module with basis $\mathcal{B}$.
- (a) Show that N is countable.
 - (b) Show that there is some countable subset $\mathcal{B}_1$ of $\mathcal{B}$ such that N is contained in the submodule, N_1 , generated by $\mathcal{B}_1$. Show also that N_1 is countable.
 - (c) Let $\overline{M} = M/N_1$. Show that $\overline{M}$ is a free $\mathbb{Z}$ -module. Deduce that if $\overline{x}$ is any nonzero element of $\overline{M}$ then there are only finitely many distinct positive integers k such that $\overline{x} = k\overline{m}$ for some $m \in M$ (depending on k).
 - (d) Let $\mathcal{S} = \{(b_1, b_2, b_3, \dots) \mid b_i = \pm i! \text{ for all } i\}$. Prove that $\mathcal{S}$ is uncountable. Deduce that there is some $s \in \mathcal{S}$ with $s \notin N_1$.
 - (e) Show that the assumption M is free leads to a contradiction: By (d) we may choose $s \in \mathcal{S}$ with $s \notin N_1$. Show that for each positive integer k there is some $m \in M$ with $\overline{s} = k\overline{m}$, contrary to (c). [Use the fact that $N \subseteq N_1$.]
25. In the construction of direct limits, Exercise 8 of Section 7.6, show that if all A_i are R -modules and the maps ρ_{ij} are R -module homomorphisms, then the direct limit $A = \varinjlim A_i$ may be given the structure of an R -module in a natural way such that the maps $\rho_i : A_i \rightarrow A$ are all R -module homomorphisms. Verify the corresponding universal property (part (e)) for R -module homomorphisms $\varphi_i : A_i \rightarrow C$ commuting with the ρ_{ij} .
26. Carry out the analysis of the preceding exercise corresponding to inverse limits to show that an inverse limit of R -modules is an R -module satisfying the appropriate universal property (cf. Exercise 10 of Section 7.6).
27. (*Free modules over noncommutative rings need not have a unique rank*) Let M be the $\mathbb{Z}$ -module $\mathbb{Z} \times \mathbb{Z} \times \dots$ of Exercise 24 and let R be its endomorphism ring, $R = \text{End}_{\mathbb{Z}}(M)$ (cf. Exercises 29 and 30 in Section 7.1). Define $\varphi_1, \varphi_2 \in R$ by

$$\varphi_1(a_1, a_2, a_3, \dots) = (a_1, a_3, a_5, \dots)$$

$$\varphi_2(a_1, a_2, a_3, \dots) = (a_2, a_4, a_6, \dots)$$

- (a) Prove that $\{\varphi_1, \varphi_2\}$ is a free basis of the left R -module R . [Define the maps ψ_1 and ψ_2 by $\psi_1(a_1, a_2, \dots) = (a_1, 0, a_2, 0, \dots)$ and $\psi_2(a_1, a_2, \dots) = (0, a_1, 0, a_2, \dots)$. Verify that $\varphi_i \psi_i = 1$, $\varphi_1 \psi_2 = 0 = \varphi_2 \psi_1$ and $\psi_1 \varphi_1 + \psi_2 \varphi_2 = 1$. Use these relations to prove that φ_1, φ_2 are independent and generate R as a left R -module.]
- (b) Use (a) to prove that $R \cong R^2$ and deduce that $R \cong R^n$ for all $n \in \mathbb{Z}^+$.

10.4 TENSOR PRODUCTS OF MODULES

In this section we study the tensor product of two modules M and N over a ring (not necessarily commutative) containing 1. Formation of the tensor product is a general construction that, loosely speaking, enables one to form another module in which one can take “products” mn of elements $m \in M$ and $n \in N$. The general construction involves various left- and right- module actions, and it is instructive, by way of motivation, to first consider an important special case: the question of “extending scalars” or “changing the base.”

Suppose that the ring R is a subring of the ring S . Throughout this section, we always assume that $1_R = 1_S$ (this ensures that S is a unital R -module).

If N is a left S -module, then N can also be naturally considered as a left R -module since the elements of R (being elements of S) act on N by assumption. The S -module axioms for N include the relations

$$(s_1 + s_2)n = s_1n + s_2n \quad \text{and} \quad s(n_1 + n_2) = sn_1 + sn_2 \quad (10.1)$$

for all $s, s_1, s_2 \in S$ and all $n, n_1, n_2 \in N$, and the relation

$$(s_1s_2)n = s_1(s_2n) \quad \text{for all } s_1, s_2 \in S, \text{ and all } n \in N. \quad (10.2)$$

A particular case of the latter relation is

$$(sr)n = s(rn) \quad \text{for all } s \in S, r \in R \text{ and } n \in N. \quad (10.2')$$

More generally, if $f: R \rightarrow S$ is a ring homomorphism from R into S with $f(1_R) = 1_S$ (for example the injection map if R is a subring of S as above) then it is easy to see that N can be considered as an R -module with $rn = f(r)n$ for $r \in R$ and $n \in N$. In this situation S can be considered as an *extension* of the ring R and the resulting R -module is said to be obtained from N by *restriction of scalars* from S to R .

Suppose now that R is a subring of S and we try to reverse this, namely we start with an R -module N and attempt to define an S -module structure on N that extends the action of R on N to an action of S on N (hence “extending the scalars” from R to S). In general this is impossible, even in the simplest situation: the ring R itself is an R -module but is usually not an S -module for the larger ring S . For example, $\mathbb{Z}$ is a $\mathbb{Z}$ -module but it cannot be made into a $\mathbb{Q}$ -module (if it could, then $\frac{1}{2} \circ 1 = z$ would be an element of $\mathbb{Z}$ with $z + z = 1$, which is impossible). Although $\mathbb{Z}$ itself cannot be made into a $\mathbb{Q}$ -module it is *contained* in a $\mathbb{Q}$ -module, namely $\mathbb{Q}$ itself. Put another way, there is an injection (also called an *embedding*) of the $\mathbb{Z}$ -module $\mathbb{Z}$ into the $\mathbb{Q}$ -module $\mathbb{Q}$ (and similarly the ring R can always be embedded as an R -submodule of the S -module S). This raises the question of whether an arbitrary R -module N can be embedded as an R -submodule of some S -module, or more generally, the question of what R -module homomorphisms exist from N to S -modules. For example, suppose N is a nontrivial *finite* abelian group, say $N = \mathbb{Z}/2\mathbb{Z}$, and consider possible $\mathbb{Z}$ -module homomorphisms (i.e., abelian group homomorphisms) of N into some $\mathbb{Q}$ -module. A $\mathbb{Q}$ -module is just a vector space over $\mathbb{Q}$ and every nonzero element in a vector space over $\mathbb{Q}$ has infinite (additive) order. Since every element of N has finite order, every element of N must map to 0 under such a homomorphism. In other words there are *no* nonzero $\mathbb{Z}$ -module homomorphisms from this N to *any* $\mathbb{Q}$ -module, much less embeddings of N identifying

N as a submodule of a $\mathbb{Q}$ -module. The two $\mathbb{Z}$ -modules $\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$ exhibit extremely different behaviors when we try to “extend scalars” from $\mathbb{Z}$ to $\mathbb{Q}$: the first module maps injectively into some $\mathbb{Q}$ -module, the second always maps to 0 in a $\mathbb{Q}$ -module.

We now construct for a general R -module N an S -module that is the “best possible” target in which to try to embed N . We shall also see that this module determines *all* of the possible R -module homomorphisms of N into S -modules, in particular determining when N is contained in some S -module (cf. Corollary 9). In the case of $R = \mathbb{Z}$ and $S = \mathbb{Q}$ this construction will give us $\mathbb{Q}$ when applied to the module $N = \mathbb{Z}$, and will give us 0 when applied to the module $N = \mathbb{Z}/2\mathbb{Z}$ (Examples 2 and 3 following Corollary 9).

If the R -module N were already an S -module then of course there is no difficulty in “extending” the scalars from R to S , so we begin the construction by returning to the basic module axioms in order to examine whether we can define “products” of the form sn , for $s \in S$ and $n \in N$. These axioms start with an abelian group N together with a map from $S \times N$ to N , where the image of the pair (s, n) is denoted by sn . It is therefore natural to consider the free $\mathbb{Z}$ -module (i.e., the free abelian group) on the set $S \times N$, i.e., the collection of all finite commuting sums of elements of the form (s_i, n_i) where $s_i \in S$ and $n_i \in N$. This is an abelian group where there are no relations between any distinct pairs (s, n) and (s', n') , i.e., no relations between the “formal products” sn , and in this abelian group the original module N has been thoroughly distinguished from the new “coefficients” from S . To satisfy the relations necessary for an S -module structure imposed in equation (1) and the compatibility relation with the action of R on N in (2'), we must take the quotient of this abelian group by the subgroup H generated by all elements of the form

$$\begin{aligned} & (s_1 + s_2, n) - (s_1, n) - (s_2, n), \\ & (s, n_1 + n_2) - (s, n_1) - (s, n_2), \text{ and} \\ & (sr, n) - (s, rn), \end{aligned} \tag{10.3}$$

for $s, s_1, s_2 \in S$, $n, n_1, n_2 \in N$ and $r \in R$, where rn in the last element refers to the R -module structure already defined on N .

The resulting quotient group is denoted by $S \otimes_R N$ (or just $S \otimes N$ if R is clear from the context) and is called the *tensor product of S and N over R* . If $s \otimes n$ denotes the coset containing (s, n) in $S \otimes_R N$ then by definition of the quotient we have forced the relations

$$\begin{aligned} & (s_1 + s_2) \otimes n = s_1 \otimes n + s_2 \otimes n, \\ & s \otimes (n_1 + n_2) = s \otimes n_1 + s \otimes n_2, \text{ and} \\ & sr \otimes n = s \otimes rn. \end{aligned} \tag{10.4}$$

The elements of $S \otimes_R N$ are called *tensors* and can be written (non-uniquely in general) as finite sums of “simple tensors” of the form $s \otimes n$ with $s \in S$, $n \in N$.

We now show that the tensor product $S \otimes_R N$ is naturally a left S -module under the action defined by

$$s \left(\sum_{\text{finite}} s_i \otimes n_i \right) = \sum_{\text{finite}} (ss_i) \otimes n_i. \tag{10.5}$$

We first check this is well defined, i.e., independent of the representation of the element of $S \otimes_R N$ as a sum of simple tensors. Note first that if s' is any element of S then

$$\begin{aligned} (s'(s_1 + s_2), n) - (s's_1, n) - (s's_2, n) &= ((s's_1 + s's_2), n) - (s's_1, n) - (s's_2, n), \\ (s's, n_1 + n_2) - (s's, n_1) - (s's, n_2), &\text{ and} \\ (s'(sr), n) - (s's, rn) &= ((s's)r, n) - (s's, rn) \end{aligned}$$

each belongs to the set of generators in (3), so in particular each lies in the subgroup H . This shows that multiplying the first entries of the generators in (3) on the left by s' gives another element of H (in fact another generator). Since any element of H is a sum of elements as in (3), it follows that for any element $\sum (s_i, n_i)$ in H also $\sum (s's_i, n_i)$ lies in H . Suppose now that $\sum s_i \otimes n_i = \sum s'_i \otimes n'_i$ are two representations for the same element in $S \otimes_R N$. Then $\sum (s_i, n_i) - \sum (s'_i, n'_i)$ is an element of H , and by what we have just seen, for any $s \in S$ also $\sum (ss_i, n_i) - \sum (ss'_i, n'_i)$ is an element of H . But this means that $\sum ss_i \otimes n_i = \sum ss'_i \otimes n'_i$ in $S \otimes_R N$, so the expression in (5) is indeed well defined.

It is now straightforward using the relations in (4) to check that the action defined in (5) makes $S \otimes_R N$ into a left S -module. For example, on the simple tensor $s_i \otimes n_i$,

$$\begin{aligned} (s + s')(s_i \otimes n_i) &= ((s + s')s_i) \otimes n_i && \text{by definition (5)} \\ &= (ss_i + s's_i) \otimes n_i \\ &= ss_i \otimes n_i + s's_i \otimes n_i && \text{by the first relation in (4)} \\ &= s(s_i \otimes n_i) + s'(s_i \otimes n_i) && \text{by definition (5).} \end{aligned}$$

The module $S \otimes_R N$ is called *the (left) S -module obtained by extension of scalars from the (left) R -module N* .

There is a natural map $\iota : N \rightarrow S \otimes_R N$ defined by $n \mapsto 1 \otimes n$ (i.e., first map $n \in N$ to the element $(1, n)$ in the free abelian group and then pass to the quotient group). Since $1 \otimes rn = r \otimes n = r(1 \otimes n)$ by (4) and (5), it is easy to check that ι is an R -module homomorphism from N to $S \otimes_R N$. Since we have passed to a quotient group, however, ι is not injective in general. Hence, while there is a natural R -module homomorphism from the original left R -module N to the left S -module $S \otimes_R N$, in general $S \otimes_R N$ need not contain (an isomorphic copy of) N . On the other hand, the relations in equation (3) were the *minimal* relations that we had to impose in order to obtain an S -module, so it is reasonable to expect that the tensor product $S \otimes_R N$ is the “best possible” S -module to serve as target for an R -module homomorphism from N . The next theorem makes this more precise by showing that any other R -module homomorphism from N factors through this one, and is referred to as the *universal property* for the tensor product $S \otimes_R N$. The analogous result for the general tensor product is given in Theorem 10.

Theorem 8. Let R be a subring of S , let N be a left R -module and let $\iota : N \rightarrow S \otimes_R N$ be the R -module homomorphism defined by $\iota(n) = 1 \otimes n$. Suppose that L is any left S -module (hence also an R -module) and that $\varphi : N \rightarrow L$ is an R -module homomorphism from N to L . Then there is a unique S -module homomorphism $\Phi : S \otimes_R N \rightarrow L$ such that φ factors through Φ , i.e., $\varphi = \Phi \circ \iota$ and the diagram

$$\begin{array}{ccc} N & \xrightarrow{\iota} & S \otimes_R N \\ & \searrow \varphi & \downarrow \Phi \\ & & L \end{array}$$

commutes. Conversely, if $\Phi : S \otimes_R N \rightarrow L$ is an S -module homomorphism then $\varphi = \Phi \circ \iota$ is an R -module homomorphism from N to L .

Proof: Suppose $\varphi : N \rightarrow L$ is an R -module homomorphism to the S -module L . By the universal property of free modules (Theorem 6 in Section 3) there is a $\mathbb{Z}$ -module homomorphism from the free $\mathbb{Z}$ -module F on the set $S \times N$ to L that sends each generator (s, n) to $s\varphi(n)$. Since φ is an R -module homomorphism, the generators of the subgroup H in equation (3) all map to zero in L . Hence this $\mathbb{Z}$ -module homomorphism factors through H , i.e., there is a well defined $\mathbb{Z}$ -module homomorphism Φ from $F/H = S \otimes_R N$ to L satisfying $\Phi(s \otimes n) = s\varphi(n)$. Moreover, on simple tensors we have

$$s'\Phi(s \otimes n) = s'(s\varphi(n)) = (s's)\varphi(n) = \Phi((s's) \otimes n) = \Phi(s'(s \otimes n)).$$

for any $s' \in S$. Since Φ is additive it follows that Φ is an S -module homomorphism, which proves the existence statement of the theorem. The module $S \otimes_R N$ is generated as an S -module by elements of the form $1 \otimes n$, so any S -module homomorphism is uniquely determined by its values on these elements. Since $\Phi(1 \otimes n) = \varphi(n)$, it follows that the S -module homomorphism Φ is uniquely determined by φ , which proves the uniqueness statement of the theorem. The converse statement is immediate.

The universal property of $S \otimes_R N$ in Theorem 8 shows that R -module homomorphisms of N into S -modules arise from S -module homomorphisms from $S \otimes_R N$. In particular this determines when it is possible to map N injectively into some S -module:

Corollary 9. Let $\iota : N \rightarrow S \otimes_R N$ be the R -module homomorphism in Theorem 8. Then $N/\ker \iota$ is the unique largest quotient of N that can be embedded in any S -module. In particular, N can be embedded as an R -submodule of some left S -module if and only if ι is injective (in which case N is isomorphic to the R -submodule $\iota(N)$ of the S -module $S \otimes_R N$).

Proof: The quotient $N/\ker \iota$ is mapped injectively (by ι) into the S -module $S \otimes_R N$. Suppose now that φ is an R -module homomorphism injecting the quotient $N/\ker \varphi$ of N into an S -module L . Then, by Theorem 8, $\ker \iota$ is mapped to 0 by φ , i.e., $\ker \iota \subseteq \ker \varphi$. Hence $N/\ker \varphi$ is a quotient of $N/\ker \iota$ (namely, the quotient by the submodule $\ker \varphi/\ker \iota$). It follows that $N/\ker \iota$ is the unique largest quotient of N that can be embedded in any S -module. The last statement in the corollary follows immediately.

Examples

- (1) For any ring R and any left R -module N we have $R \otimes_R N \cong N$ (so “extending scalars from R to R ” does not change the module). This follows by taking φ to be the identity map from N to itself (and $S = R$) in Theorem 8: ι is then an isomorphism with inverse isomorphism given by Φ . In particular, if A is any abelian group (i.e., a $\mathbb{Z}$ -module), then $\mathbb{Z} \otimes_{\mathbb{Z}} A = A$.
- (2) Let $R = \mathbb{Z}$, $S = \mathbb{Q}$ and let A be a finite abelian group of order n . In this case the $\mathbb{Q}$ -module $\mathbb{Q} \otimes_{\mathbb{Z}} A$ obtained by extension of scalars from the $\mathbb{Z}$ -module A is 0. To see this, observe first that in any tensor product $1 \otimes 0 = 1 \otimes (0 + 0) = 1 \otimes 0 + 1 \otimes 0$, by the second relation in (4), so

$$1 \otimes 0 = 0.$$

Now, for any simple tensor $q \otimes a$ we can write the rational number q as $(q/n)n$. Then since $na = 0$ in A by Lagrange’s Theorem, we have

$$q \otimes a = \left(\frac{q}{n} \cdot n\right) \otimes a = \frac{q}{n} \otimes (na) = (q/n) \otimes 0 = (q/n)(1 \otimes 0) = 0.$$

It follows that $\mathbb{Q} \otimes_{\mathbb{Z}} A = 0$. In particular, the map $\iota : A \rightarrow S \otimes_R A$ is the zero map. By Theorem 8, we see again that any homomorphism of a finite abelian group into a rational vector space is the zero map. In particular, if A is nontrivial, then the original $\mathbb{Z}$ -module A is not contained in the $\mathbb{Q}$ -module obtained by extension of scalars.

- (3) *Extension of scalars for free modules:* If $N \cong R^n$ is a free module of rank n over R then $S \otimes_R N \cong S^n$ is a free module of rank n over S . We shall prove this shortly (Corollary 18) when we discuss tensor products of direct sums. For example, $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}^n \cong \mathbb{Q}^n$. In this case the module obtained by extension of scalars contains (an isomorphic copy of) the original R -module N . For example, $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}^n \cong \mathbb{Q}^n$ and $\mathbb{Z}^n$ is a subgroup of the abelian group $\mathbb{Q}^n$.
- (4) *Extension of scalars for vector spaces:* As a special case of the previous example, let F be a subfield of the field K and let V be an n -dimensional vector space over F (i.e., $V \cong F^n$). Then $K \otimes_F V \cong K^n$ is a vector space over the larger field K of the same dimension, and the original vector space V is contained in $K \otimes_F V$ as an F -vector subspace.
- (5) *Induced modules for finite groups:* Let R be a commutative ring with 1, let G be a finite group and let H be a subgroup of G . As in Section 7.2 we may form the group ring RG and its subring RH . For any RH -module N define the *induced module* $RG \otimes_{RH} N$. In this way we obtain an RG -module for each RH -module N . We shall study properties of induced modules and some of their important applications to group theory in Chapters 17 and 19.

The general tensor product construction follows along the same lines as the extension of scalars above, but before describing it we make two observations from this special case. The first is that the construction of $S \otimes_R N$ as an *abelian group* involved only the elements in equation (3), which in turn only required S to be a *right* R -module and N to be a *left* R -module. In a similar way we shall construct an *abelian group* $M \otimes_R N$ for any *right* R -module M and any *left* R -module N . The second observation is that the S -module structure on $S \otimes_R N$ defined by equation (5) required only a *left* S -module structure on S together with a “compatibility relation”

$$s'(sr) = (s's)r \quad \text{for } s, s' \in S, r \in R,$$

between this left S -module structure and the right R -module structure on S (this was needed in order to deduce that (5) was well defined). We first consider the general construction of $M \otimes_R N$ as an abelian group, after which we shall return to the question of when this abelian group can be given a module structure.

Suppose then that N is a left R -module and that M is a right R -module. The quotient of the free $\mathbb{Z}$ -module on the set $M \times N$ by the subgroup generated by all elements of the form

$$\begin{aligned} (m_1 + m_2, n) - (m_1, n) - (m_2, n), \\ (m, n_1 + n_2) - (m, n_1) - (m, n_2), \text{ and} \\ (mr, n) - (m, rn), \end{aligned} \tag{10.6}$$

for $m, m_1, m_2 \in M, n, n_1, n_2 \in N$ and $r \in R$ is an abelian group, denoted by $M \otimes_R N$, or simply $M \otimes N$ if the ring R is clear from the context, and is called the *tensor product of M and N over R* . The elements of $M \otimes_R N$ are called *tensors*, and the coset, $m \otimes n$, of (m, n) in $M \otimes_R N$ is called a *simple tensor*. We have the relations

$$\begin{aligned} (m_1 + m_2) \otimes n &= m_1 \otimes n + m_2 \otimes n, \\ m \otimes (n_1 + n_2) &= m \otimes n_1 + m \otimes n_2, \text{ and} \\ mr \otimes n &= m \otimes rn. \end{aligned} \tag{10.7}$$

Every tensor can be written (non-uniquely in general) as a finite sum of simple tensors.

Remark: We emphasize that care must be taken when working with tensors, since each $m \otimes n$ represents a *coset* in some quotient group, and so we may have $m \otimes n = m' \otimes n'$ where $m \neq m'$ or $n \neq n'$. More generally, an element of $M \otimes N$ may be expressible in many different ways as a sum of simple tensors. In particular, care must be taken when defining maps from $M \otimes_R N$ to another group or module, since a map from $M \otimes N$ which is described on the generators $m \otimes n$ in terms of m and n is not well defined unless it is shown to be independent of the particular choice of $m \otimes n$ as a coset representative.

Another point where care must be exercised is in reference to the element $m \otimes n$ when the modules M and N or the ring R are not clear from the context. The first two examples of extension of scalars give an instance where M is a submodule of a larger module M' , and for some $m \in M$ and $n \in N$ we have $m \otimes n = 0$ in $M' \otimes_R N$ but $m \otimes n$ is *nonzero* in $M \otimes_R N$. This is possible because the symbol “ $m \otimes n$ ” represents different cosets, hence possibly different elements, in the two tensor products. In particular, these two examples show that $M \otimes_R N$ need not be a subgroup of $M' \otimes_R N$ even when M is a submodule of M' (cf. also Exercise 2).

Mapping $M \times N$ to the free $\mathbb{Z}$ -module on $M \times N$ and then passing to the quotient defines a map $\iota : M \times N \rightarrow M \otimes_R N$ with $\iota(m, n) = m \otimes n$. This map is in general not a group homomorphism, but it is additive in both m and n separately and satisfies $\iota(mr, n) = mr \otimes n = m \otimes rn = \iota(m, rn)$. Such maps are given a name:

Definition. Let M be a right R -module, let N be a left R -module and let L be an abelian group (written additively). A map $\varphi : M \times N \rightarrow L$ is called *R -balanced* or *middle linear with respect to R* if

$$\varphi(m_1 + m_2, n) = \varphi(m_1, n) + \varphi(m_2, n)$$

$$\varphi(m, n_1 + n_2) = \varphi(m, n_1) + \varphi(m, n_2)$$

$$\varphi(m, rn) = \varphi(mr, n)$$

for all $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, and $r \in R$.

With this terminology, it follows immediately from the relations in (7) that the map $\iota : M \times N \rightarrow M \otimes_R N$ is R -balanced. The next theorem proves the **extremely useful universal property of the tensor product** with respect to balanced maps.

Theorem 10. Suppose R is a ring with 1, M is a right R -module, and N is a left R -module. Let $M \otimes_R N$ be the tensor product of M and N over R and let $\iota : M \times N \rightarrow M \otimes_R N$ be the R -balanced map defined above.

- (1) If $\Phi : M \otimes_R N \rightarrow L$ is any group homomorphism from $M \otimes_R N$ to an abelian group L then the composite map $\varphi = \Phi \circ \iota$ is an R -balanced map from $M \times N$ to L .
- (2) Conversely, suppose L is an abelian group and $\varphi : M \times N \rightarrow L$ is any R -balanced map. Then there is a unique group homomorphism $\Phi : M \otimes_R N \rightarrow L$ such that φ factors through ι , i.e., $\varphi = \Phi \circ \iota$ as in (1).

Equivalently, the correspondence $\varphi \leftrightarrow \Phi$ in the commutative diagram

$$\begin{array}{ccc} M \times N & \xrightarrow{\iota} & M \otimes_R N \\ & \searrow \varphi & \downarrow \Phi \\ & & L \end{array}$$

establishes a bijection

$$\left\{ \begin{array}{l} R\text{-balanced maps} \\ \varphi : M \times N \rightarrow L \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} \text{group homomorphisms} \\ \Phi : M \otimes_R N \rightarrow L \end{array} \right\}.$$

Proof: The proof of (1) is immediate from the properties of ι above. For (2), the map φ defines a unique $\mathbb{Z}$ -module homomorphism $\tilde{\varphi}$ from the free group on $M \times N$ to L (Theorem 6 in Section 3) such that $\tilde{\varphi}(m, n) = \varphi(m, n) \in L$. Since φ is R -balanced, $\tilde{\varphi}$ maps each of the elements in equation (6) to 0; for example

$$\tilde{\varphi}((mr, n) - (m, rn)) = \varphi(mr, n) - \varphi(m, rn) = 0.$$

It follows that the kernel of $\tilde{\varphi}$ contains the subgroup generated by these elements, hence $\tilde{\varphi}$ induces a homomorphism Φ on the quotient group $M \otimes_R N$ to L . By definition we then have

$$\Phi(m \otimes n) = \tilde{\varphi}(m, n) = \varphi(m, n),$$

i.e., $\varphi = \Phi \circ \iota$. The homomorphism Φ is uniquely determined by this equation since the elements $m \otimes n$ generate $M \otimes_R N$ as an abelian group. This completes the proof.

Theorem 10 is extremely useful in defining homomorphisms on $M \otimes_R N$ since it replaces the often tedious check that maps defined on simple tensors $m \otimes n$ are well defined with a check that a related map defined on ordered pairs (m, n) is balanced.

The first consequence of the universal property in Theorem 10 is a characterization of the tensor product $M \otimes_R N$ as an abelian group:

Corollary 11. Suppose D is an abelian group and $\iota' : M \times N \rightarrow D$ is an R -balanced map such that

- (i) the image of ι' generates D as an abelian group, and
- (ii) every R -balanced map defined on $M \times N$ factors through ι' as in Theorem 10.

Then there is an isomorphism $f : M \otimes_R N \cong D$ of abelian groups with $\iota' = f \circ \iota$.

Proof: Since $\iota' : M \times N \rightarrow D$ is a balanced map, the universal property in (2) of Theorem 10 implies there is a (unique) homomorphism $f : M \otimes_R N \rightarrow D$ with $\iota' = f \circ \iota$. In particular $\iota'(m, n) = f(m \otimes n)$ for every $m \in M, n \in N$. By the first assumption on ι' , these elements generate D as an abelian group, so f is a surjective map. Now, the balanced map $\iota : M \times N \rightarrow M \otimes_R N$ together with the second assumption on ι' implies there is a (unique) homomorphism $g : D \rightarrow M \otimes_R N$ with $\iota = g \circ \iota'$. Then $m \otimes n = (g \circ f)(m \otimes n)$. Since the simple tensors $m \otimes n$ generate $M \otimes_R N$, it follows that $g \circ f$ is the identity map on $M \otimes_R N$ and so f is injective, hence an isomorphism. This establishes the corollary.

We now return to the question of giving the abelian group $M \otimes_R N$ a *module* structure. As we observed in the special case of extending scalars from R to S for the R -module N , the S -module structure on $S \otimes_R N$ required only a left S -module structure on S together with the compatibility relation $s'(sr) = (s's)r$ for $s, s' \in S$ and $r \in R$. In this special case this relation was simply a consequence of the associative law in the ring S . To obtain an S -module structure on $M \otimes_R N$ more generally we impose a similar structure on M :

Definition. Let R and S be any rings with 1. An abelian group M is called an (S, R) -*bimodule* if M is a left S -module, a right R -module, and $s(mr) = (sm)r$ for all $s \in S, r \in R$ and $m \in M$.

Examples

- (1) Any ring S is an (S, R) -bimodule for any subring R with $1_R = 1_S$ by the associativity of the multiplication in S . More generally, if $f : R \rightarrow S$ is any ring homomorphism with $f(1_R) = 1_S$ then S can be considered as a right R -module with the action $s \cdot r = sf(r)$, and with respect to this action S becomes an (S, R) -bimodule.
- (2) Let I be an ideal (two-sided) in the ring R . Then the quotient ring R/I is an $(R/I, R)$ -bimodule. This is easy to see directly and is also a special case of the previous example (with respect to the canonical projection homomorphism $R \rightarrow R/I$).
- (3) Suppose that R is a commutative ring. Then a left (respectively, right) R -module M can always be given the structure of a right (respectively, left) R -module by defining $mr = rm$ (respectively, $rm = mr$), for all $m \in M$ and $r \in R$, and this makes M into

an (R, R) -bimodule. Hence every module (right or left) over a commutative ring R has at least one natural (R, R) -bimodule structure.

- (4) Suppose that M is a left S -module and R is a subring contained in the *center* of S (for example, if S is commutative). Then in particular R is commutative so M can be given a right R -module structure as in the previous example. Then for any $s \in S$, $r \in R$ and $m \in M$ by definition of the right action of R we have

$$(sm)r = r(sm) = (rs)m = (sr)m = s(rm) = s(mr)$$

(note that we have used the fact that r commutes with s in the middle equality). Hence M is an (S, R) -bimodule with respect to this definition of the right action of R .

Since the situation in Example 3 occurs so frequently, we give this bimodule structure a name:

Definition. Suppose M is a left (or right) R -module over the commutative ring R . Then the (R, R) -bimodule structure on M defined by letting the left and right R -actions coincide, i.e., $mr = rm$ for all $m \in M$ and $r \in R$, will be called the *standard* R -module structure on M .

Suppose now that N is a left R -module and M is an (S, R) -bimodule. Then just as in the example of extension of scalars the (S, R) -bimodule structure on M implies that

$$s \left(\sum_{\text{finite}} m_i \otimes n_i \right) = \sum_{\text{finite}} (sm_i) \otimes n_i \quad (10.8)$$

gives a well defined action of S under which $M \otimes_R N$ is a left S -module. Note that Theorem 10 may be used to give an alternate proof that (8) is well defined, replacing the direct calculations on the relations defining the tensor product with the easier check that a map is R -balanced, as follows. It is very easy to see that for each fixed $s \in S$ the map $(m, n) \mapsto sm \otimes n$ is an R -balanced map from $M \times N$ to $M \otimes_R N$. By Theorem 10 there is a well defined group homomorphism λ_s from $M \otimes_R N$ to itself such that $\lambda_s(m \otimes n) = sm \otimes n$. Since the right side of (8) is then $\lambda_s(\sum m_i \otimes n_i)$, the fact that λ_s is well defined shows that this expression is indeed independent of the representation of the tensor $\sum m_i \otimes n_i$ as a sum of simple tensors. Because λ_s is additive, equation (8) holds.

By a completely parallel argument, if M is a right R -module and N is an (R, S) -bimodule then the tensor product $M \otimes_R N$ has the structure of a right S -module, where $(\sum m_i \otimes n_i)s = \sum m_i \otimes (n_i s)$.

Before giving some more examples of tensor products it is worthwhile to highlight one frequently encountered special case of the previous discussion, namely the case when M and N are two left modules over a commutative ring R and $S = R$ (in some works on tensor products this is the only case considered). Then the standard R -module structure on M defined previously gives M the structure of an (R, R) -bimodule, so in this case the tensor product $M \otimes_R N$ always has the structure of a left R -module.

The corresponding map $\iota : M \times N \rightarrow M \otimes_R N$ maps $M \times N$ into an R -module and is additive in each factor. Since $r(m \otimes n) = rm \otimes n = mr \otimes n = m \otimes rn$ it also satisfies

$$r\iota(m, n) = \iota(rm, n) = \iota(m, rn).$$

Such maps are given a name:

Definition. Let R be a commutative ring with 1 and let M , N , and L be left R -modules. The map $\varphi : M \times N \rightarrow L$ is called R -bilinear if it is R -linear in each factor, i.e., if

$$\begin{aligned}\varphi(r_1 m_1 + r_2 m_2, n) &= r_1 \varphi(m_1, n) + r_2 \varphi(m_2, n), \quad \text{and} \\ \varphi(m, r_1 n_1 + r_2 n_2) &= r_1 \varphi(m, n_1) + r_2 \varphi(m, n_2)\end{aligned}$$

for all $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$ and $r_1, r_2 \in R$.

With this terminology Theorem 10 gives

Corollary 12. Suppose R is a commutative ring. Let M and N be two left R -modules and let $M \otimes_R N$ be the tensor product of M and N over R , where M is given the standard R -module structure. Then $M \otimes_R N$ is a left R -module with

$$r(m \otimes n) = (rm) \otimes n = (mr) \otimes n = m \otimes (rn),$$

and the map $\iota : M \times N \rightarrow M \otimes_R N$ with $\iota(m, n) = m \otimes n$ is an R -bilinear map. If L is any left R -module then there is a bijection

$$\left\{ \begin{array}{l} R\text{-bilinear maps} \\ \varphi : M \times N \rightarrow L \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{l} R\text{-module homomorphisms} \\ \Phi : M \otimes_R N \rightarrow L \end{array} \right\}$$

where the correspondence between φ and Φ is given by the commutative diagram

$$\begin{array}{ccc} M \times N & \xrightarrow{\iota} & M \otimes_R N \\ & \searrow \varphi & \downarrow \Phi \\ & & L \end{array}$$

Proof: We have shown $M \otimes_R N$ is an R -module and that ι is bilinear. It remains only to check that in the bijective correspondence in Theorem 10 the bilinear maps correspond with the R -module homomorphisms. If $\varphi : M \times N \rightarrow L$ is bilinear then it is an R -balanced map, so the corresponding $\Phi : M \otimes_R N \rightarrow L$ is a group homomorphism. Moreover, on simple tensors $\Phi((rm) \otimes n) = \varphi(rm, n) = r\varphi(m, n) = r\Phi(m \otimes n)$, where the middle equality holds because φ is R -linear in the first variable. Since Φ is additive this extends to sums of simple tensors to show Φ is an R -module homomorphism. Conversely, if Φ is an R -module homomorphism it is an exercise to see that the corresponding balanced map φ is bilinear.

Examples

- (1) In any tensor product $M \otimes_R N$ we have $m \otimes 0 = m \otimes (0 + 0) = (m \otimes 0) + (m \otimes 0)$, so $m \otimes 0 = 0$. Likewise $0 \otimes n = 0$.
- (2) We have $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z} = 0$, since $3a = a$ for $a \in \mathbb{Z}/2\mathbb{Z}$ so that

$$a \otimes b = 3a \otimes b = a \otimes 3b = a \otimes 0 = 0$$

and every simple tensor is reduced to 0. In particular $1 \otimes 1 = 0$. It follows that there are no nonzero balanced (or bilinear) maps from $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ to any abelian group.

On the other hand, consider the tensor product $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$, which is generated as an abelian group by the elements $0 \otimes 0 = 1 \otimes 0 = 0 \otimes 1 = 0$ and $1 \otimes 1$. In this case $1 \otimes 1 \neq 0$ since, for example, the map $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ defined by $(a, b) \mapsto ab$ is clearly nonzero and linear in both a and b . Since $2(1 \otimes 1) = 2 \otimes 1 = 0 \otimes 1 = 0$, the element $1 \otimes 1$ is of order 2. Hence $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$.

- (3) In general,

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z},$$

where d is the g.c.d. of the integers m and n . To see this, observe first that

$$a \otimes b = a \otimes (b \cdot 1) = (ab) \otimes 1 = ab(1 \otimes 1),$$

from which it follows that $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$ is a cyclic group with $1 \otimes 1$ as generator. Since $m(1 \otimes 1) = m \otimes 1 = 0 \otimes 1 = 0$ and similarly $n(1 \otimes 1) = 1 \otimes n = 0$, we have $d(1 \otimes 1) = 0$, so the cyclic group has order dividing d . The map $\varphi : \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/d\mathbb{Z}$ defined by $\varphi(a \bmod m, b \bmod n) = ab \bmod d$ is well defined since d divides both m and n . It is clearly $\mathbb{Z}$ -bilinear. The induced map $\Phi : \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/d\mathbb{Z}$ from Corollary 12 maps $1 \otimes 1$ to the element $1 \in \mathbb{Z}/d\mathbb{Z}$, which is an element of order d . In particular $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$ has order at least d . Hence $1 \otimes 1$ is an element of order d and Φ gives an isomorphism $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$.

- (4) In $\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z}$ a simple tensor has the form $(a/b \bmod \mathbb{Z}) \otimes (c/d \bmod \mathbb{Z})$ for some rational numbers a/b and c/d . Then

$$\begin{aligned} \left(\frac{a}{b} \bmod \mathbb{Z}\right) \otimes \left(\frac{c}{d} \bmod \mathbb{Z}\right) &= d\left(\frac{a}{bd} \bmod \mathbb{Z}\right) \otimes \left(\frac{c}{d} \bmod \mathbb{Z}\right) \\ &= \left(\frac{a}{bd} \bmod \mathbb{Z}\right) \otimes d\left(\frac{c}{d} \bmod \mathbb{Z}\right) = \left(\frac{a}{bd} \bmod \mathbb{Z}\right) \otimes 0 = 0 \end{aligned}$$

and so

$$\mathbb{Q}/\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} = 0.$$

In a similar way, $A \otimes_{\mathbb{Z}} B = 0$ for any *divisible* abelian group A and *torsion* abelian group B (an abelian group in which every element has finite order). For example

$$\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}/\mathbb{Z} = 0.$$

- (5) The structure of a tensor product can vary considerably depending on the ring over which the tensors are taken. For example $\mathbb{Q} \otimes_{\mathbb{Q}} \mathbb{Q}$ and $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ are isomorphic as left $\mathbb{Q}$ -modules (both are one dimensional vector spaces over $\mathbb{Q}$) — cf. the exercises. On the other hand we shall see at the end of this section that $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$ and $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ are not isomorphic $\mathbb{C}$ -modules (the former is a 1-dimensional vector space over $\mathbb{C}$ and the latter is 2-dimensional over $\mathbb{C}$).
- (6) *General extension of scalars or change of base:* Let $f : R \rightarrow S$ be a ring homomorphism with $f(1_R) = 1_S$. Then $s \cdot r = sf(r)$ gives S the structure of a right R -module with respect to which S is an (S, R) -bimodule. Then for any left R -module N , the resulting tensor product $S \otimes_R N$ is a left S -module obtained by *changing the base* from R to S . This gives a slight generalization of the notion of extension of scalars (where R was a subring of S).
- (7) Let $f : R \rightarrow S$ be a ring homomorphism as in the preceding example. Then we have $S \otimes_R R \cong S$ as left S -modules, as follows. The map $\varphi : S \times R \rightarrow S$ defined by $(s, r) \mapsto sr$ (where $sr = sf(r)$ by definition of the right R -action on S), is an R -balanced map, as is easily checked. For example,

$$\varphi(s_1 + s_2, r) = (s_1 + s_2)r = s_1r + s_2r = \varphi(s_1, r) + \varphi(s_2, r)$$

and

$$\varphi(sr, r') = (sr)r' = s(rr') = \varphi(s, rr').$$

By Theorem 10 we have an associated group homomorphism $\Phi : S \otimes_R R \rightarrow S$ with $\Phi(s \otimes r) = sr$. Since $\Phi(s'(s \otimes r)) = \Phi(s's \otimes r) = s'sr = s'\Phi(s \otimes r)$, it follows that Φ is also an S -module homomorphism. The map $\Phi' : S \rightarrow S \otimes_R R$ with $s \mapsto s \otimes 1$ is an S -module homomorphism that is inverse to Φ because $\Phi \circ \Phi'(s) = \Phi(s \otimes 1) = s$ gives $\Phi\Phi' = 1$, and

$$\Phi' \circ \Phi(s \otimes r) = \Phi'(sr) = sr \otimes 1 = s \otimes r$$

shows that $\Phi'\Phi$ is the identity on simple tensors, hence $\Phi'\Phi = 1$.

- (8) Let R be a ring (not necessarily commutative), let I be a two sided ideal in R , and let N be a left R -module. Then as previously mentioned, R/I is an $(R/I, R)$ -bimodule, so the tensor product $R/I \otimes_R N$ is a left R/I -module. This is an example of “extension of scalars” with respect to the natural projection homomorphism $R \rightarrow R/I$.

Define

$$IN = \left\{ \sum_{\text{finite}} a_i n_i \mid a_i \in I, n_i \in N \right\},$$

which is easily seen to be a left R -submodule of N (cf. Exercise 5, Section 1). Then

$$(R/I) \otimes_R N \cong N/IN,$$

as left R -modules, as follows. The tensor product is generated as an abelian group by the simple tensors $(r \bmod I) \otimes n = r(1 \otimes n)$ for $r \in R$ and $n \in N$ (viewing the R/I -module tensor product as an R -module on which I acts trivially). Hence the elements $1 \otimes n$ generate $(R/I) \otimes_R N$ as an R/I -module. The map $N \rightarrow (R/I) \otimes_R N$ defined by $n \mapsto 1 \otimes n$ is a left R -module homomorphism and, by the previous observation, is surjective. Under this map $a_i n_i$ with $a_i \in I$ and $n_i \in N$ maps to $1 \otimes a_i n_i = a_i \otimes n_i = 0$, and so IN is contained in the kernel. This induces a surjective R -module homomorphism $f : N/IN \rightarrow (R/I) \otimes_R N$ with $f(n \bmod I) = 1 \otimes n$. We show f is an isomorphism by exhibiting its inverse. The map $(R/I) \times N \rightarrow N/IN$ defined by mapping $(r \bmod I, n)$ to $(rn \bmod IN)$ is well defined and easily checked to be R -balanced. It follows by Theorem 10 that there is an associated group homomorphism $g : (R/I) \otimes_R N \rightarrow N/IN$ with $g((r \bmod I) \otimes n) = rn \bmod IN$. As usual, $fg = 1$ and $gf = 1$, so f is a bijection and $(R/I) \otimes_R N \cong N/IN$, as claimed.

As an example, let $R = \mathbb{Z}$ with ideal $I = m\mathbb{Z}$ and let N be the $\mathbb{Z}$ -module $\mathbb{Z}/n\mathbb{Z}$. Then $IN = m(\mathbb{Z}/n\mathbb{Z}) = (m\mathbb{Z} + n\mathbb{Z})/n\mathbb{Z} = d\mathbb{Z}/n\mathbb{Z}$ where d is the g.c.d. of m and n . Then $N/IN \cong \mathbb{Z}/d\mathbb{Z}$ and we recover the isomorphism $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$ of Example 3 above.

We now establish some of the basic properties of tensor products. Note the frequent application of Theorem 10 to establish the existence of homomorphisms.

Theorem 13. (*The “Tensor Product” of Two Homomorphisms*) Let M, M' be right R -modules, let N, N' be left R -modules, and suppose $\varphi : M \rightarrow M'$ and $\psi : N \rightarrow N'$ are R -module homomorphisms.

- (1) There is a unique group homomorphism, denoted by $\varphi \otimes \psi$, mapping $M \otimes_R N$ into $M' \otimes_R N'$ such that $(\varphi \otimes \psi)(m \otimes n) = \varphi(m) \otimes \psi(n)$ for all $m \in M$ and $n \in N$.

- (2) If M, M' are also (S, R) -bimodules for some ring S and φ is also an S -module homomorphism, then $\varphi \otimes \psi$ is a homomorphism of left S -modules. In particular, if R is commutative then $\varphi \otimes \psi$ is always an R -module homomorphism for the standard R -module structures.
- (3) If $\lambda : M' \rightarrow M''$ and $\mu : N' \rightarrow N''$ are R -module homomorphisms then $(\lambda \otimes \mu) \circ (\varphi \otimes \psi) = (\lambda \circ \varphi) \otimes (\mu \circ \psi)$.

Proof: The map $(m, n) \mapsto \varphi(m) \otimes \psi(n)$ from $M \times N$ to $M' \otimes_R N'$ is clearly R -balanced, so (1) follows immediately from Theorem 10.

In (2) the definition of the (left) action of S on M together with the assumption that φ is an S -module homomorphism imply that on simple tensors

$$(\varphi \otimes \psi)(s(m \otimes n)) = (\varphi \otimes \psi)(sm \otimes n) = \varphi(sm) \otimes \psi(n) = s\varphi(m) \otimes \psi(n).$$

Since $\varphi \otimes \psi$ is additive, this extends to sums of simple tensors to show that $\varphi \otimes \psi$ is an S -module homomorphism. This gives (2).

The uniqueness condition in Theorem 10 implies (3), which completes the proof.

The next result shows that we may write $M \otimes N \otimes L$, or more generally, an n -fold tensor product $M_1 \otimes M_2 \otimes \cdots \otimes M_n$, unambiguously whenever it is defined.

Theorem 14. (*Associativity of the Tensor Product*) Suppose M is a right R -module, N is an (R, T) -bimodule, and L is a left T -module. Then there is a unique isomorphism

$$(M \otimes_R N) \otimes_T L \cong M \otimes_R (N \otimes_T L)$$

of abelian groups such that $(m \otimes n) \otimes l \mapsto m \otimes (n \otimes l)$. If M is an (S, R) -bimodule, then this is an isomorphism of S -modules.

Proof: Note first that the (R, T) -bimodule structure on N makes $M \otimes_R N$ into a right T -module and $N \otimes_T L$ into a left R -module, so both sides of the isomorphism are well defined. For each fixed $l \in L$, the mapping $(m, n) \mapsto m \otimes (n \otimes l)$ is R -balanced, so by Theorem 10 there is a homomorphism $M \otimes_R N \rightarrow M \otimes_R (N \otimes_T L)$ with $m \otimes n \mapsto m \otimes (n \otimes l)$. This shows that the map from $(M \otimes_R N) \times L$ to $M \otimes_R (N \otimes_T L)$ given by $(m \otimes n, l) \mapsto m \otimes (n \otimes l)$ is well defined. Since it is easily seen to be T -balanced, another application of Theorem 10 implies that it induces a homomorphism $(M \otimes_R N) \otimes_T L \rightarrow M \otimes_R (N \otimes_T L)$ such that $(m \otimes n) \otimes l \mapsto m \otimes (n \otimes l)$. In a similar way we can construct a homomorphism in the opposite direction that is inverse to this one. This proves the group isomorphism.

Assume in addition M is an (S, R) -bimodule. Then for $s \in S$ and $t \in T$ we have

$$s((m \otimes n)t) = s(m \otimes nt) = sm \otimes nt = (sm \otimes n)t = (s(m \otimes n))t$$

so that $M \otimes_R N$ is an (S, T) -bimodule. Hence $(M \otimes_R N) \otimes_T L$ is a left S -module. Since $N \otimes_T L$ is a left R -module, also $M \otimes_R (N \otimes_T L)$ is a left S -module. The group isomorphism just established is easily seen to be a homomorphism of left S -modules by the same arguments used in previous proofs: it is additive and is S -linear on simple tensors since $s((m \otimes n) \otimes l) = s(m \otimes n) \otimes l = (sm \otimes n) \otimes l$ maps to the element $sm \otimes (n \otimes l) = s(m \otimes (n \otimes l))$. The proof is complete.

Corollary 15. Suppose R is commutative and M, N , and L are left R -modules. Then

$$(M \otimes N) \otimes L \cong M \otimes (N \otimes L)$$

as R -modules for the standard R -module structures on M, N and L .

There is a natural extension of the notion of a bilinear map:

Definition. Let R be a commutative ring with 1 and let $M_1, M_2, \dots, M_n$ and L be R -modules with the standard R -module structures. A map $\varphi : M_1 \times \dots \times M_n \rightarrow L$ is called *n -multilinear over R* (or simply *multilinear* if n and R are clear from the context) if it is an R -module homomorphism in each component when the other component entries are kept constant, i.e., for each i

$$\begin{aligned} \varphi(m_1, \dots, m_{i-1}, rm_i + r'm'_i, m_{i+1}, \dots, m_n) \\ = r\varphi(m_1, \dots, m_i, \dots, m_n) + r'\varphi(m_1, \dots, m'_i, \dots, m_n) \end{aligned}$$

for all $m_i, m'_i \in M_i$ and $r, r' \in R$. When $n = 2$ (respectively, 3) one says φ is *bilinear* (respectively *trilinear*) rather than 2-multilinear (or 3-multilinear).

One may construct the n -fold tensor product $M_1 \otimes M_2 \otimes \dots \otimes M_n$ from first principles and prove its analogous universal property with respect to multilinear maps from $M_1 \times \dots \times M_n$ to L . By the previous theorem and corollary, however, an n -fold tensor product may be obtained unambiguously by iterating the tensor product of pairs of modules since any bracketing of $M_1 \otimes \dots \otimes M_n$ into tensor products of pairs gives an isomorphic R -module. The universal property of the tensor product of a pair of modules in Theorem 10 and Corollary 12 then implies that multilinear maps factor uniquely through the R -module $M_1 \otimes \dots \otimes M_n$, i.e., this tensor product is the universal object with respect to multilinear functions:

Corollary 16. Let R be a commutative ring and let $M_1, \dots, M_n, L$ be R -modules. Let $M_1 \otimes M_2 \otimes \dots \otimes M_n$ denote any bracketing of the tensor product of these modules and let

$$\iota : M_1 \times \dots \times M_n \rightarrow M_1 \otimes \dots \otimes M_n$$

be the map defined by $\iota(m_1, \dots, m_n) = m_1 \otimes \dots \otimes m_n$. Then

- (1) for every R -module homomorphism $\Phi : M_1 \otimes \dots \otimes M_n \rightarrow L$ the map $\varphi = \Phi \circ \iota$ is n -multilinear from $M_1 \times \dots \times M_n$ to L , and
- (2) if $\varphi : M_1 \times \dots \times M_n \rightarrow L$ is an n -multilinear map then there is a unique R -module homomorphism $\Phi : M_1 \otimes \dots \otimes M_n \rightarrow L$ such that $\varphi = \Phi \circ \iota$.

Hence there is a bijection

$$\left\{ \begin{array}{c} n\text{-multilinear maps} \\ \varphi : M_1 \times \dots \times M_n \rightarrow L \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} R\text{-module homomorphisms} \\ \Phi : M_1 \otimes \dots \otimes M_n \rightarrow L \end{array} \right\}$$

with respect to which the following diagram commutes:

$$\begin{array}{ccc} M \times \dots \times M_n & \xrightarrow{\iota} & M \otimes \dots \otimes M_n \\ & \searrow \varphi & \downarrow \Phi \\ & & L \end{array}$$

We have already seen examples where $M_1 \otimes_R N$ is not contained in $M \otimes_R N$ even when M_1 is an R -submodule of M . The next result shows in particular that (an isomorphic copy of) $M_1 \otimes_R N$ is contained in $M \otimes_R N$ if M_1 is an R -module *direct summand* of M .

Theorem 17. (Tensor Products of Direct Sums) Let M, M' be right R -modules and let N, N' be left R -modules. Then there are unique group isomorphisms

$$(M \oplus M') \otimes_R N \cong (M \otimes_R N) \oplus (M' \otimes_R N)$$

$$M \otimes_R (N \oplus N') \cong (M \otimes_R N) \oplus (M \otimes_R N')$$

such that $(m, m') \otimes n \mapsto (m \otimes n, m' \otimes n)$ and $m \otimes (n, n') \mapsto (m \otimes n, m \otimes n')$ respectively. If M, M' are also (S, R) -bimodules, then these are isomorphisms of left S -modules. In particular, if R is commutative, these are isomorphisms of R -modules.

Proof: The map $(M \oplus M') \times N \rightarrow (M \otimes_R N) \oplus (M' \otimes_R N)$ defined by $((m, m'), n) \mapsto (m \otimes n, m' \otimes n)$ is well defined since m and m' in $M \oplus M'$ are uniquely defined in the direct sum. The map is clearly R -balanced, so induces a homomorphism f from $(M \oplus M') \otimes N$ to $(M \otimes_R N) \oplus (M' \otimes_R N)$ with

$$f((m, m') \otimes n) = (m \otimes n, m' \otimes n).$$

In the other direction, the R -balanced maps $M \times N \rightarrow (M \oplus M') \otimes_R N$ and $M' \times N \rightarrow (M \oplus M') \otimes_R N$ given by $(m, n) \mapsto (m, 0) \otimes n$ and $(m', n) \mapsto (0, m') \otimes n$, respectively, define homomorphisms from $M \otimes_R N$ and $M' \otimes_R N$ to $(M \oplus M') \otimes_R N$. These in turn give a homomorphism g from the direct sum $(M \otimes_R N) \oplus (M' \otimes_R N)$ to $(M \oplus M') \otimes_R N$ with

$$g((m \otimes n_1, m' \otimes n_2)) = (m, 0) \otimes n_1 + (0, m') \otimes n_2.$$

An easy check shows that f and g are inverse homomorphisms and are S -module isomorphisms when M and M' are (S, R) -bimodules. This completes the proof.

The previous theorem clearly extends by induction to any finite direct sum of R -modules. The corresponding result is also true for arbitrary direct sums. For example

$$M \otimes (\oplus_{i \in I} N_i) \cong \oplus_{i \in I} (M \otimes N_i),$$

where I is any index set (cf. the exercises). This result is referred to by saying that *tensor products commute with direct sums*.

Corollary 18. (Extension of Scalars for Free Modules) The module obtained from the free R -module $N \cong R^n$ by extension of scalars from R to S is the free S -module S^n , i.e.,

$$S \otimes_R R^n \cong S^n$$

as left S -modules.

Proof: This follows immediately from Theorem 17 and the isomorphism $S \otimes_R R \cong S$ proved in Example 7 previously.

Corollary 19. Let R be a commutative ring and let $M \cong R^s$ and $N \cong R^t$ be free R -modules with bases $m_1, \dots, m_s$ and $n_1, \dots, n_t$, respectively. Then $M \otimes_R N$ is a free R -module of rank st , with basis $m_i \otimes n_j$, $1 \leq i \leq s$ and $1 \leq j \leq t$, i.e.,

$$R^s \otimes_R R^t \cong R^{st}.$$

Remark: More generally, the tensor product of two free modules of arbitrary rank over a commutative ring is free (cf. the exercises).

Proof: This follows easily from Theorem 17 and the first example following Corollary 9.

Proposition 20. Suppose R is a commutative ring and M, N are left R -modules, considered with the standard R -module structures. Then there is a unique R -module isomorphism

$$M \otimes_R N \cong N \otimes_R M$$

mapping $m \otimes n$ to $n \otimes m$.

Proof: The map $M \times N \rightarrow N \otimes M$ defined by $(m, n) \mapsto n \otimes m$ is R -balanced. Hence it induces a unique homomorphism f from $M \otimes N$ to $N \otimes M$ with $f(m \otimes n) = n \otimes m$. Similarly, we have a unique homomorphism g from $N \otimes M$ to $M \otimes N$ with $g(n \otimes m) = m \otimes n$ giving the inverse of f , and both maps are easily seen to be R -module isomorphisms.

Remark: When $M = N$ it is not in general true that $a \otimes b = b \otimes a$ for $a, b \in M$. We shall study “symmetric tensors” in Section 11.6.

We end this section by showing that the tensor product of R -algebras is again an R -algebra.

Proposition 21. Let R be a commutative ring and let A and B be R -algebras. Then the multiplication $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$ is well defined and makes $A \otimes_R B$ into an R -algebra.

Proof: Note first that the definition of an R -algebra shows that

$$r(a \otimes b) = ra \otimes b = ar \otimes b = a \otimes rb = a \otimes br = (a \otimes b)r$$

for every $r \in R, a \in A$ and $b \in B$. To show that $A \otimes B$ is an R -algebra the main task is, as usual, showing that the specified multiplication is well defined. One way to proceed is to use two applications of Corollary 16, as follows. The map $\varphi : A \times B \times A \times B \rightarrow A \otimes B$ defined by $\varphi(a, b, a', b') = aa' \otimes bb'$ is multilinear over R . For example,

$$\begin{aligned} f(a, r_1 b_1 + r_2 b_2, a', b') &= aa' \otimes (r_1 b_1 + r_2 b_2) b' \\ &= aa' \otimes r_1 b_1 b' + aa' \otimes r_2 b_2 b' \\ &= r_1 f(a, b_1, a', b') + r_2 f(a, b_2, a', b'). \end{aligned}$$

By Corollary 16, there is a corresponding R -module homomorphism Φ from $A \otimes B \otimes A \otimes B$ to $A \otimes B$ with $\Phi(a \otimes b \otimes a' \otimes b') = aa' \otimes bb'$. Viewing $A \otimes B \otimes A \otimes B$ as $(A \otimes B) \otimes (A \otimes B)$, we can apply Corollary 16 once more to obtain a well defined R -bilinear mapping φ' from $(A \otimes B) \times (A \otimes B)$ to $A \otimes B$ with $\varphi'(a \otimes b, a' \otimes b') = aa' \otimes bb'$. This shows that the multiplication is indeed well defined (and also that it satisfies the distributive laws). It is now a simple matter (left to the exercises) to check that with this multiplication $A \otimes B$ is an R -algebra.

Example

The tensor product $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ is free of rank 4 as a module over $\mathbb{R}$ with basis given by $e_1 = 1 \otimes 1$, $e_2 = 1 \otimes i$, $e_3 = i \otimes 1$, and $e_4 = i \otimes i$ (by Corollary 19). By Proposition 21, this tensor product is also a (commutative) ring with $e_1 = 1$, and, for example,

$$e_4^2 = (i \otimes i)(i \otimes i) = i^2 \otimes i^2 = (-1) \otimes (-1) = (-1)(-1) \otimes 1 = 1.$$

Then $(e_4 - 1)(e_4 + 1) = 0$, so $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ is not an integral domain.

The ring $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ is an $\mathbb{R}$ -algebra and the left and right $\mathbb{R}$ -actions are the same: $xr = rx$ for every $r \in \mathbb{R}$ and $x \in \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$. The ring $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ has a structure of a left $\mathbb{C}$ -module because the first $\mathbb{C}$ is a $(\mathbb{C}, \mathbb{R})$ -bimodule. It also has a right $\mathbb{C}$ -module structure because the second $\mathbb{C}$ is an $(\mathbb{R}, \mathbb{C})$ -bimodule. For example,

$$i \cdot e_1 = i \cdot (1 \otimes 1) = (i \cdot 1) \otimes 1 = i \otimes 1 = e_3$$

and

$$e_1 \cdot i = (1 \otimes 1) \cdot i = 1 \otimes (1 \cdot i) = 1 \otimes i = e_2.$$

This example also shows that even when the rings involved are commutative there may be natural left and right module structures (over some ring) that are not the same.

EXERCISES

Let R be a ring with 1.

1. Let $f : R \rightarrow S$ be a ring homomorphism from the ring R to the ring S with $f(1_R) = 1_S$. Verify the details that $sr = sf(r)$ defines a right R -action on S under which S is an (S, R) -bimodule.
2. Show that the element " $2 \otimes 1$ " is 0 in $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ but is nonzero in $2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$.
3. Show that $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ and $\mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$ are both left $\mathbb{R}$ -modules but are not isomorphic as $\mathbb{R}$ -modules.
4. Show that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ and $\mathbb{Q} \otimes_{\mathbb{Q}} \mathbb{Q}$ are isomorphic left $\mathbb{Q}$ -modules. [Show they are both 1-dimensional vector spaces over $\mathbb{Q}$.]
5. Let A be a finite abelian group of order n and let p^k be the largest power of the prime p dividing n . Prove that $\mathbb{Z}/p^k\mathbb{Z} \otimes_{\mathbb{Z}} A$ is isomorphic to the Sylow p -subgroup of A .
6. If R is any integral domain with quotient field Q , prove that $(Q/R) \otimes_R (Q/R) = 0$.
7. If R is any integral domain with quotient field Q and N is a left R -module, prove that every element of the tensor product $Q \otimes_R N$ can be written as a simple tensor of the form $(1/d) \otimes n$ for some nonzero $d \in R$ and some $n \in N$.
8. Suppose R is an integral domain with quotient field Q and let N be any R -module. Let $U = R^\times$ be the set of nonzero elements in R and define $U^{-1}N$ to be the set of equivalence classes of ordered pairs of elements (u, n) with $u \in U$ and $n \in N$ under the equivalence relation $(u, n) \sim (u', n')$ if and only if $u'n = un'$ in N .

- (a) Prove that $U^{-1}N$ is an abelian group under the addition defined by $\overline{(u_1, n_1)} + \overline{(u_2, n_2)} = \overline{(u_1u_2, u_2n_1 + u_1n_2)}$. Prove that $r\overline{(u, n)} = \overline{(u, rn)}$ defines an action of R on $U^{-1}N$ making it into an R -module. [This is an example of *localization* considered in general in Section 4 of Chapter 15, cf. also Section 5 in Chapter 7.]
- (b) Show that the map from $Q \times N$ to $U^{-1}N$ defined by sending $(a/b, n)$ to $\overline{(b, an)}$ for $a \in R, b \in U, n \in N$, is an R -balanced map, so induces a homomorphism f from $Q \otimes_R N$ to $U^{-1}N$. Show that the map g from $U^{-1}N$ to $Q \otimes_R N$ defined by $g(\overline{(u, n)}) = (1/u) \otimes n$ is well defined and is an inverse homomorphism to f . Conclude that $Q \otimes_R N \cong U^{-1}N$ as R -modules.
- (c) Conclude from (b) that $(1/d) \otimes n$ is 0 in $Q \otimes_R N$ if and only if $rn = 0$ for some nonzero $r \in R$.
- (d) If A is an abelian group, show that $Q \otimes_{\mathbb{Z}} A = 0$ if and only if A is a torsion abelian group (i.e., every element of A has finite order).
9. Suppose R is an integral domain with quotient field Q and let N be any R -module. Let $Q \otimes_R N$ be the module obtained from N by extension of scalars from R to Q . Prove that the kernel of the R -module homomorphism $\iota : N \rightarrow Q \otimes_R N$ is the torsion submodule of N (cf. Exercise 8 in Section 1). [Use the previous exercise.]
10. Suppose R is commutative and $N \cong R^n$ is a free R -module of rank n with R -module basis $e_1, \dots, e_n$.
- (a) For any nonzero R -module M show that every element of $M \otimes N$ can be written uniquely in the form $\sum_{i=1}^n m_i \otimes e_i$ where $m_i \in M$. Deduce that if $\sum_{i=1}^n m_i \otimes e_i = 0$ in $M \otimes N$ then $m_i = 0$ for $i = 1, \dots, n$.
- (b) Show that if $\sum m_i \otimes n_i = 0$ in $M \otimes N$ where the n_i are merely assumed to be R -linearly independent then it is not necessarily true that all the m_i are 0. [Consider $R = \mathbb{Z}, n = 1, M = \mathbb{Z}/2\mathbb{Z}$, and the element $1 \otimes 2$.]
11. Let $\{e_1, e_2\}$ be a basis of $V = \mathbb{R}^2$. Show that the element $e_1 \otimes e_2 + e_2 \otimes e_1$ in $V \otimes_{\mathbb{R}} V$ cannot be written as a simple tensor $v \otimes w$ for any $v, w \in \mathbb{R}^2$.
12. Let V be a vector space over the field F and let v, v' be nonzero elements of V . Prove that $v \otimes v' = v' \otimes v$ in $V \otimes_F V$ if and only if $v = av'$ for some $a \in F$.
13. Prove that the usual dot product of vectors defined by letting $(a_1, \dots, a_n) \cdot (b_1, \dots, b_n)$ be $a_1b_1 + \dots + a_nb_n$ is a bilinear map from $\mathbb{R}^n \times \mathbb{R}^n$ to $\mathbb{R}$.
14. Let I be an arbitrary nonempty index set and for each $i \in I$ let N_i be a left R -module. Let M be a right R -module. Prove the group isomorphism: $M \otimes (\oplus_{i \in I} N_i) \cong \oplus_{i \in I} (M \otimes N_i)$, where the direct sum of an arbitrary collection of modules is defined in Exercise 20, Section 3. [Use the same argument as for the direct sum of two modules, taking care to note where the direct sum hypothesis is needed — cf. the next exercise.]
15. Show that tensor products do not commute with direct products in general. [Consider the extension of scalars from $\mathbb{Z}$ to $\mathbb{Q}$ of the direct product of the modules $M_i = \mathbb{Z}/2^i\mathbb{Z}$, $i = 1, 2, \dots$]
16. Suppose R is commutative and let I and J be ideals of R , so R/I and R/J are naturally R -modules.
- (a) Prove that every element of $R/I \otimes_R R/J$ can be written as a simple tensor of the form $(1 \bmod I) \otimes (r \bmod J)$.
- (b) Prove that there is an R -module isomorphism $R/I \otimes_R R/J \cong R/(I + J)$ mapping $(r \bmod I) \otimes (r' \bmod J)$ to $rr' \bmod (I + J)$.
17. Let $I = (2, x)$ be the ideal generated by 2 and x in the ring $R = \mathbb{Z}[x]$. The ring $\mathbb{Z}/2\mathbb{Z} = R/I$ is naturally an R -module annihilated by both 2 and x .

(a) Show that the map $\varphi : I \times I \rightarrow \mathbb{Z}/2\mathbb{Z}$ defined by

$$\varphi(a_0 + a_1x + \cdots + a_nx^n, b_0 + b_1x + \cdots + b_mx^m) = \frac{a_0}{2}b_1 \bmod 2$$

is R -bilinear.

(b) Show that there is an R -module homomorphism from $I \otimes_R I \rightarrow \mathbb{Z}/2\mathbb{Z}$ mapping

$$p(x) \otimes q(x) \text{ to } \frac{p(0)}{2}q'(0) \text{ where } q' \text{ denotes the usual polynomial derivative of } q.$$

(c) Show that $2 \otimes x \neq x \otimes 2$ in $I \otimes_R I$.

18. Suppose I is a principal ideal in the integral domain R . Prove that the R -module $I \otimes_R I$ has no nonzero torsion elements (i.e., $rm = 0$ with $0 \neq r \in R$ and $m \in I \otimes_R I$ implies that $m = 0$).

19. Let $I = (2, x)$ be the ideal generated by 2 and x in the ring $R = \mathbb{Z}[x]$ as in Exercise 17. Show that the nonzero element $2 \otimes x - x \otimes 2$ in $I \otimes_R I$ is a torsion element. Show in fact that $2 \otimes x - x \otimes 2$ is annihilated by both 2 and x and that the submodule of $I \otimes_R I$ generated by $2 \otimes x - x \otimes 2$ is isomorphic to R/I .

20. Let $I = (2, x)$ be the ideal generated by 2 and x in the ring $R = \mathbb{Z}[x]$. Show that the element $2 \otimes 2 + x \otimes x$ in $I \otimes_R I$ is not a simple tensor, i.e., cannot be written as $a \otimes b$ for some $a, b \in I$.

21. Suppose R is commutative and let I and J be ideals of R .

(a) Show there is a surjective R -module homomorphism from $I \otimes_R J$ to the product ideal IJ mapping $i \otimes j$ to the element ij .

(b) Give an example to show that the map in (a) need not be injective (cf. Exercise 17).

22. Suppose that M is a left and a right R -module such that $rm = mr$ for all $r \in R$ and $m \in M$. Show that the elements r_1r_2 and r_2r_1 act the same on M for every $r_1, r_2 \in R$. (This explains why the assumption that R is commutative in the definition of an R -algebra is a fairly natural one.)

23. Verify the details that the multiplication in Proposition 19 makes $A \otimes_R B$ into an R -algebra.

24. Prove that the extension of scalars from $\mathbb{Z}$ to the Gaussian integers $\mathbb{Z}[i]$ of the ring $\mathbb{R}$ is isomorphic to $\mathbb{C}$ as a ring: $\mathbb{Z}[i] \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{C}$ as rings.

25. Let R be a subring of the commutative ring S and let x be an indeterminate over S . Prove that $S[x]$ and $S \otimes_R R[x]$ are isomorphic as S -algebras.

26. Let S be a commutative ring containing R (with $1_S = 1_R$) and let $x_1, \dots, x_n$ be independent indeterminates over the ring S . Show that for every ideal I in the polynomial ring $R[x_1, \dots, x_n]$ that $S \otimes_R (R[x_1, \dots, x_n]/I) \cong S[x_1, \dots, x_n]/IS[x_1, \dots, x_n]$ as S -algebras.

The next exercise shows the ring $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ introduced at the end of this section is isomorphic to $\mathbb{C} \times \mathbb{C}$. One may also prove this via Exercise 26 and Proposition 16 in Section 9.5, since $\mathbb{C} \cong \mathbb{R}[x]/(x^2 + 1)$. The ring $\mathbb{C} \times \mathbb{C}$ is also discussed in Exercise 23 of Section 1.

27. (a) Write down a formula for the multiplication of two elements $a \cdot 1 + b \cdot e_2 + c \cdot e_3 + d \cdot e_4$ and $a' \cdot 1 + b' \cdot e_2 + c' \cdot e_3 + d' \cdot e_4$ in the example $A = \mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ following Proposition 21 (where $1 = 1 \otimes 1$ is the identity of A).

(b) Let $\epsilon_1 = \frac{1}{2}(1 \otimes 1 + i \otimes i)$ and $\epsilon_2 = \frac{1}{2}(1 \otimes 1 - i \otimes i)$. Show that $\epsilon_1 \epsilon_2 = 0$, $\epsilon_1 + \epsilon_2 = 1$, and $\epsilon_j^2 = \epsilon_j$ for $j = 1, 2$ (ϵ_1 and ϵ_2 are called *orthogonal idempotents* in A). Deduce that A is isomorphic as a ring to the direct product of two principal ideals: $A \cong A\epsilon_1 \times A\epsilon_2$ (cf. Exercise 1, Section 7.6).

(c) Prove that the map $\varphi : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C} \times \mathbb{C}$ by $\varphi(z_1, z_2) = (z_1 z_2, z_1 \overline{z_2})$, where $\overline{z_2}$ denotes the complex conjugate of z_2 , is an $\mathbb{R}$ -bilinear map.

- (d) Let Φ be the $\mathbb{R}$ -module homomorphism from A to $\mathbb{C} \times \mathbb{C}$ obtained from φ in (c). Show that $\Phi(\epsilon_1) = (0, 1)$ and $\Phi(\epsilon_2) = (1, 0)$. Show also that Φ is $\mathbb{C}$ -linear, where the action of $\mathbb{C}$ is on the left tensor factor in A and on both factors in $\mathbb{C} \times \mathbb{C}$. Deduce that Φ is surjective. Show that Φ is a $\mathbb{C}$ -algebra isomorphism.

10.5 EXACT SEQUENCES—PROJECTIVE, INJECTIVE, AND FLAT MODULES

One of the fundamental results for studying the structure of an algebraic object B (e.g., a group, a ring, or a module) is the First Isomorphism Theorem, which relates the subobjects of B (the normal subgroups, the ideals, or the submodules, respectively) with the possible homomorphic images of B . We have already seen many examples applying this theorem to understand the structure of B from an understanding of its “smaller” constituents—for example in analyzing the structure of the dihedral group D_8 by determining its center and the resulting quotient by the center.

In most of these examples we began *first* with a given B and then determined some of its basic properties by constructing a homomorphism φ (often given implicitly by the specification of $\ker \varphi \subseteq B$) and examining both $\ker \varphi$ and the resulting quotient $B/\ker \varphi$. We now consider in some greater detail the reverse situation, namely whether we may *first* specify the “smaller constituents.” More precisely, we consider whether, given two modules A and C , there exists a module B containing (an isomorphic copy of) A such that the resulting quotient module B/A is isomorphic to C —in which case B is said to be an *extension of C by A* . It is then natural to ask how many such B exist for a given A and C , and the extent to which properties of B are determined by the corresponding properties of A and C . There are, of course, analogous problems in the contexts of groups and rings. This is the *extension problem* first discussed (for groups) in Section 3.4; in this section we shall be primarily concerned with left modules over a ring R , making note where necessary of the modifications required for some other structures, notably noncommutative groups. As in the previous section, throughout this section all rings contain a 1.

We first introduce a very convenient notation. To say that A is isomorphic to a submodule of B , is to say that there is an injective homomorphism $\psi : A \rightarrow B$ (so then $A \cong \psi(A) \subseteq B$). To say that C is isomorphic to the resulting quotient is to say that there is a surjective homomorphism $\varphi : B \rightarrow C$ with $\ker \varphi = \psi(A)$. In particular this gives us a pair of homomorphisms:

$$A \xrightarrow{\psi} B \xrightarrow{\varphi} C$$

with image $\psi = \ker \varphi$. A pair of homomorphisms with this property is given a name:

Definition.

- (1) The pair of homomorphisms $X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z$ is said to be *exact* (at Y) if $\operatorname{im} \alpha = \ker \beta$.
- (2) A sequence $\cdots \rightarrow X_{n-1} \rightarrow X_n \rightarrow X_{n+1} \rightarrow \cdots$ of homomorphisms is said to be an *exact sequence* if it is exact at every X_n between a pair of homomorphisms.

With this terminology, the pair of homomorphisms $A \xrightarrow{\psi} B \xrightarrow{\varphi} C$ above is exact at B . We can also use this terminology to express the fact that for these maps ψ is injective and φ is surjective:

Proposition 22. Let A , B and C be R -modules over some ring R . Then

- (1) The sequence $0 \rightarrow A \xrightarrow{\psi} B$ is exact (at A) if and only if ψ is injective.
- (2) The sequence $B \xrightarrow{\varphi} C \rightarrow 0$ is exact (at C) if and only if φ is surjective.

Proof: The (uniquely defined) homomorphism $0 \rightarrow A$ has image 0 in A . This will be the kernel of ψ if and only if ψ is injective. Similarly, the kernel of the (uniquely defined) zero homomorphism $C \rightarrow 0$ is all of C , which is the image of φ if and only if φ is surjective.

Corollary 23. The sequence $0 \rightarrow A \xrightarrow{\psi} B \xrightarrow{\varphi} C \rightarrow 0$ is exact if and only if ψ is injective, φ is surjective, and $\text{image } \psi = \ker \varphi$, i.e., B is an extension of C by A .

Definition. The exact sequence $0 \rightarrow A \xrightarrow{\psi} B \xrightarrow{\varphi} C \rightarrow 0$ is called a *short exact sequence*.

In terms of this notation, the extension problem can be stated succinctly as follows: given modules A and C , determine all the short exact sequences

$$0 \rightarrow A \xrightarrow{\psi} B \xrightarrow{\varphi} C \rightarrow 0. \quad (10.9)$$

We shall see below that the exact sequence notation is also extremely convenient for analyzing the extent to which properties of A and C determine the corresponding properties of B . If A , B and C are groups written multiplicatively, the sequence (9) will be written

$$1 \rightarrow A \xrightarrow{\psi} B \xrightarrow{\varphi} C \rightarrow 1 \quad (10.9')$$

where 1 denotes the trivial group. Both Proposition 22 and Corollary 23 are valid with the obvious notational changes.

Note that any exact sequence can be written as a succession of short exact sequences since to say $X \xrightarrow{\alpha} Y \xrightarrow{\beta} Z$ is exact at Y is the same as saying that the sequence $0 \rightarrow \alpha(X) \rightarrow Y \rightarrow Y/\ker \beta \rightarrow 0$ is a short exact sequence.

Examples

- (1) Given modules A and C we can always form their direct sum $B = A \oplus C$ and the sequence

$$0 \rightarrow A \xrightarrow{\iota} A \oplus C \xrightarrow{\pi} C \rightarrow 0$$

where $\iota(a) = (a, 0)$ and $\pi(a, c) = c$ is a short exact sequence. In particular, it follows that there always exists at least one extension of C by A .

- (2) As a special case of the previous example, consider the two $\mathbb{Z}$ -modules $A = \mathbb{Z}$ and $C = \mathbb{Z}/n\mathbb{Z}$:

$$0 \rightarrow \mathbb{Z} \xrightarrow{\iota} \mathbb{Z} \oplus (\mathbb{Z}/n\mathbb{Z}) \xrightarrow{\varphi} \mathbb{Z}/n\mathbb{Z} \rightarrow 0,$$

giving one extension of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$.

Another extension of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$ is given by the short exact sequence

$$0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \rightarrow 0$$

where n denotes the map $x \mapsto nx$ given by multiplication by n , and π denotes the natural projection. Note that the modules in the middle of the previous two exact sequences are not isomorphic even though the respective “A” and “C” terms are isomorphic. Thus there are (at least) two “essentially different” or “inequivalent” ways of extending $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$.

- (3) If $\varphi : B \rightarrow C$ is any homomorphism we may form an exact sequence:

$$0 \rightarrow \ker \varphi \xrightarrow{\iota} B \xrightarrow{\varphi} \text{image } \varphi \rightarrow 0$$

where ι is the inclusion map. In particular, if φ is surjective, the sequence $\varphi : B \rightarrow C$ may be extended to a short exact sequence with $A = \ker \varphi$.

- (4) One particularly important instance of the preceding example is when M is an R -module and S is a set of generators for M . Let $F(S)$ be the free R -module on S . Then

$$0 \rightarrow K \xrightarrow{\iota} F(S) \xrightarrow{\varphi} M \rightarrow 0$$

is the short exact sequence where φ is the unique R -module homomorphism which is the identity on S (cf. Theorem 6) and $K = \ker \varphi$.

More generally, when M is any group (possibly non-abelian) the above short exact sequence (with 1's at the ends, if M is written multiplicatively) describes a *presentation* of M , where K is the normal subgroup of $F(S)$ generated by the *relations* defining M (cf. Section 6.3).

- (5) Two “inequivalent” extensions G of the Klein 4-group by the cyclic group Z_2 of order two are

$$\begin{aligned} 1 &\rightarrow Z_2 \xrightarrow{\iota} D_8 \xrightarrow{\varphi} Z_2 \times Z_2 \rightarrow 1, \text{ and} \\ 1 &\rightarrow Z_2 \xrightarrow{\iota} Q_8 \xrightarrow{\varphi} Z_2 \times Z_2 \rightarrow 1, \end{aligned}$$

where in each case ι maps Z_2 injectively into the center of G (recall that both D_8 and Q_8 have centers of order two), and φ is the natural projection of G onto $G/Z(G)$.

Two other inequivalent extensions G of the Klein 4-group by Z_2 occur when G is either of the abelian groups $Z_2 \times Z_2 \times Z_2$ or $Z_2 \times Z_4$ for appropriate maps ι and φ .

Examples 2 and 5 above show that, for a fixed A and C , in general there may be several extensions of C by A . To distinguish different extensions we define the notion of a homomorphism (and isomorphism) between two exact sequences. Recall first that a diagram involving various homomorphisms is said to *commute* if any compositions of homomorphisms with the same starting and ending points are equal, i.e., the composite map defined by following a path of homomorphisms in the diagram depends only on the starting and ending points and not on the choice of the path taken.

Definition. Let $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ and $0 \rightarrow A' \rightarrow B' \rightarrow C' \rightarrow 0$ be two short exact sequences of modules.

- (1) A *homomorphism of short exact sequences* is a triple α, β, γ of module homomorphisms such that the following diagram commutes:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0 \end{array}$$

The homomorphism is an *isomorphism of short exact sequences* if α, β, γ are all isomorphisms, in which case the extensions B and B' are said to be *isomorphic extensions*.

- (2) The two exact sequences are called *equivalent* if $A = A', C = C'$, and there is an isomorphism between them as in (1) that is the identity maps on A and C (i.e., α and γ are the identity). In this case the corresponding extensions B and B' are said to be *equivalent extensions*.

If B and B' are isomorphic extensions then in particular B and B' are isomorphic as R -modules, but more is true: there is an R -module isomorphism between B and B' that restricts to an isomorphism from A to A' and induces an isomorphism on the quotients C and C' . For a given A and C the condition that two extensions B and B' of C by A are equivalent is stronger still: there must exist an R -module isomorphism between B and B' that restricts to the *identity* map on A and induces the *identity* map on C . The notion of isomorphic extensions measures how many different extensions of C by A there are, allowing for C and A to be changed by an isomorphism. The notion of equivalent extensions measures how many different extensions of C by A there are when A and C are rigidly fixed.

Homomorphisms and isomorphisms between short exact sequences of multiplicative groups (9') are defined similarly.

It is an easy exercise to see that the composition of homomorphisms of short exact sequences is also a homomorphism. Likewise, if the triple α, β, γ is an isomorphism (or equivalence) then $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$ is an isomorphism (equivalence, respectively) in the reverse direction. It follows that “isomorphism” (or equivalence) is an equivalence relation on any set of short exact sequences.

Examples

- (1) Let m and n be integers greater than 1. Assume n divides m and let $k = m/n$. Define a map from the exact sequence of $\mathbb{Z}$ -modules in Example 2 of the preceding set of examples:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Z} & \xrightarrow{n} & \mathbb{Z} & \xrightarrow{\pi} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \\ 0 & \longrightarrow & \mathbb{Z}/k\mathbb{Z} & \xrightarrow{\iota} & \mathbb{Z}/m\mathbb{Z} & \xrightarrow{\pi'} & \mathbb{Z}/n\mathbb{Z} & \longrightarrow & 0 \end{array}$$

where α and β are the natural projections, γ is the identity map, ι maps $a \bmod k$ to $na \bmod m$, and π' is the natural projection of $\mathbb{Z}/m\mathbb{Z}$ onto its quotient $(\mathbb{Z}/m\mathbb{Z})/(n\mathbb{Z}/m\mathbb{Z})$

(which is isomorphic to $\mathbb{Z}/n\mathbb{Z}$). One easily checks that this is a homomorphism of short exact sequences.

- (2) If again $0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \rightarrow 0$ is the short exact sequence of $\mathbb{Z}$ -modules defined previously, map each module to itself by $x \mapsto -x$. This triple of homomorphisms gives an isomorphism of the exact sequence with itself. This isomorphism is not an equivalence of sequences since it is not the identity on the first $\mathbb{Z}$.
- (3) The short exact sequences in Examples 1 and 2 following Corollary 23 are not isomorphic—the extension modules are not isomorphic $\mathbb{Z}$ -modules (abelian groups). Likewise the two extensions, D_8 and Q_8 , in Example 5 of the same set are not isomorphic (hence not equivalent), even though the two end terms “A” and “C” are the same for both sequences.
- (4) Consider the maps

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \xrightarrow{\psi} & \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} & \xrightarrow{\varphi} & \mathbb{Z}/2\mathbb{Z} \longrightarrow 0 \\
 & & \downarrow \text{id} & & \downarrow \beta & & \downarrow \text{id} \\
 0 & \longrightarrow & \mathbb{Z}/2\mathbb{Z} & \xrightarrow{\psi'} & \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} & \xrightarrow{\varphi'} & \mathbb{Z}/2\mathbb{Z} \longrightarrow 0
 \end{array}$$

where ψ maps $\mathbb{Z}/2\mathbb{Z}$ injectively into the first component of the direct sum and φ projects the direct sum onto its second component. Also ψ' embeds $\mathbb{Z}/2\mathbb{Z}$ into the *second* component of the direct sum and φ' projects the direct sum onto its *first* component. If β maps the direct sum $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ to itself by interchanging the two factors, then this diagram is seen to commute, hence giving an equivalence of the two exact sequences that is not the identity isomorphism.

- (5) We exhibit two isomorphic but inequivalent $\mathbb{Z}$ -module extensions. For $i = 1, 2$ define

$$0 \rightarrow \mathbb{Z}/2\mathbb{Z} \xrightarrow{\psi} \mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \xrightarrow{\varphi_i} \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \rightarrow 0$$

where $\psi : 1 \mapsto (2, 0)$ in both sequences, φ_1 is defined by $\varphi_1(a \bmod 4, b \bmod 2) = (a \bmod 2, b \bmod 2)$, and $\varphi_2(a \bmod 4, b \bmod 2) = (b \bmod 2, a \bmod 2)$. It is easy to see that the resulting two sequences are both short exact sequences.

An evident isomorphism between these two exact sequences is provided by the triple of maps $\text{id}, \text{id}, \gamma$, where $\gamma : \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ is the map $\gamma((c, d)) = (d, c)$ that interchanges the two direct factors.

We now check that these two isomorphic sequences are *not equivalent*, as follows. Since $\varphi_1(0, 1) = (0, 1)$, any equivalence, $\text{id}, \beta, \text{id}$, from the first sequence to the second must map $(0, 1) \in \mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ to either $(1, 0)$ or $(3, 0)$ in $\mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, since these are the two possible elements mapping to $(0, 1)$ by φ_2 . This is impossible, however, since the isomorphism β cannot send an element of order 2 to an element of order 4.

Put another way, equivalences involving the same extension module B are automorphisms of B that restrict to the identity on both $\psi(A)$ and $B/\psi(A)$. Any such automorphism of $B = \mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ must fix the coset $(0, 1) + \psi(A)$ since this is the unique nonidentity coset containing elements of order 2. Thus maps which send this coset to different elements in C give inequivalent extensions. In particular, there is yet a third inequivalent extension involving the same modules $A = \mathbb{Z}/2\mathbb{Z}$, $B = \mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$ and $C = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, that maps the coset $(0, 1) + \psi(A)$ to the element $(1, 1) \in \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$.

By similar reasoning there are three inequivalent but isomorphic group extensions of $\mathbb{Z}_2 \times \mathbb{Z}_2$ by $\mathbb{Z}_2$ with $B \cong D_8$ (cf. the exercises).

The homomorphisms α, β, γ in a homomorphism of short exact sequences are not independent. The next result gives some relations among these three homomorphisms.

Proposition 24. (*The Short Five Lemma*) Let α, β, γ be a homomorphism of short exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ & & \downarrow \alpha & & \downarrow \beta & & \downarrow \gamma & & \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0 \end{array}$$

- (1) If α and γ are injective then so is β .
- (2) If α and γ are surjective then so is β .
- (3) If α and γ are isomorphisms then so is β (and then the two sequences are isomorphic).

Remark: These results hold also for short exact sequences of (possibly non-abelian) groups (as the proof demonstrates).

Proof: We shall prove (1), leaving the proof of (2) as an exercise (and (3) follows immediately from (1) and (2)). Suppose then that α and γ are injective and suppose $b \in B$ with $\beta(b) = 0$. Let $\psi : A \rightarrow B$ and $\varphi : B \rightarrow C$ denote the homomorphisms in the first short exact sequence. Since $\beta(b) = 0$, it follows in particular that the image of $\beta(b)$ in the quotient C' is also 0. By the commutativity of the diagram this implies that $\gamma(\varphi(b)) = 0$, and since γ is assumed injective, we obtain $\varphi(b) = 0$, i.e., b is in the kernel of φ . By the exactness of the first sequence, this means that b is in the image of ψ , i.e., $b = \psi(a)$ for some $a \in A$. Then, again by the commutativity of the diagram, the image of $\alpha(a)$ in B' is the same as $\beta(\psi(a)) = \beta(b) = 0$. But α and the map from A' to B' are injective by assumption, and it follows that $a = 0$. Finally, $b = \psi(a) = \psi(0) = 0$ and we see that β is indeed injective.

We have already seen that there is always at least one extension of a module C by A , namely the direct sum $B = A \oplus C$. In this case the module B contains a submodule C' isomorphic to C (namely $C' = 0 \oplus C$) as well as the submodule A , and this submodule complement to A “splits” B into a direct sum. In the case of groups the existence of a subgroup complement C' to a normal subgroup in B implies that B is a semidirect product (cf. Section 5 in Chapter 5). The fact that B is a direct sum in the context of modules is a reflection of the fact that the underlying group structure in this case is *abelian*; for abelian groups semidirect products are direct products. In either case the corresponding short exact sequence is said to “split”:

Definition.

- (1) Let R be a ring and let $0 \rightarrow A \xrightarrow{\psi} B \xrightarrow{\varphi} C \rightarrow 0$ be a short exact sequence of R -modules. The sequence is said to be *split* if there is an R -module complement to $\psi(A)$ in B . In this case, up to isomorphism, $B = A \oplus C$ (more precisely, $B = \psi(A) \oplus C'$ for some submodule C' , and C' is mapped isomorphically onto C by φ : $\varphi(C') \cong C$).

- (2) If $1 \rightarrow A \xrightarrow{\psi} B \xrightarrow{\varphi} C \rightarrow 1$ is a short exact sequence of groups, then the sequence is said to be *split* if there is a subgroup complement to $\psi(A)$ in B . In this case, up to isomorphism, $B = A \rtimes C$ (more precisely, $B = \psi(A) \rtimes C'$ for some subgroup C' , and C' is mapped isomorphically onto C by φ : $\varphi(C') \cong C$).

In either case the extension B is said to be a *split extension* of C by A .

The question of whether an extension splits is the question of the existence of a complement to $\psi(A)$ in B isomorphic (by φ) to C , so the notion of a split extension may equivalently be phrased in the language of homomorphisms:

Proposition 25. The short exact sequence $0 \rightarrow A \xrightarrow{\psi} B \xrightarrow{\varphi} C \rightarrow 0$ of R -modules is split if and only if there is an R -module homomorphism $\mu : C \rightarrow B$ such that $\varphi \circ \mu$ is the identity map on C . Similarly, the short exact sequence $1 \rightarrow A \xrightarrow{\psi} B \xrightarrow{\varphi} C \rightarrow 1$ of groups is split if and only if there is a group homomorphism $\mu : C \rightarrow B$ such that $\varphi \circ \mu$ is the identity map on C .

Proof: This follows directly from the definitions: if μ is given define $C' = \mu(C) \subseteq B$ and if C' is given define $\mu = \varphi^{-1} : C \cong C' \subseteq B$.

Definition. With notation as in Proposition 25, any set map $\mu : C \rightarrow B$ such that $\varphi \circ \mu = \text{id}$ is called a *section* of φ . If μ is a *homomorphism* as in Proposition 25 then μ is called a *splitting homomorphism* for the sequence.

Note that a section of φ is nothing more than a choice of coset representatives in B for the quotient $B/\ker \varphi \cong C$. A section is a (splitting) homomorphism if this set of coset representatives forms a *submodule* (respectively, *subgroup*) in B , in which case this submodule (respectively, subgroup) gives a complement to $\psi(A)$ in B .

Examples

- (1) The split short exact sequence $0 \rightarrow A \xrightarrow{\iota} A \oplus C \xrightarrow{\pi} C \rightarrow 0$ has the evident splitting homomorphism $\mu(c) = (0, c)$.
- (2) The extension $0 \rightarrow \mathbb{Z} \xrightarrow{\iota} \mathbb{Z} \oplus (\mathbb{Z}/n\mathbb{Z}) \xrightarrow{\varphi} \mathbb{Z}/n\mathbb{Z} \rightarrow 0$, of $\mathbb{Z}/n\mathbb{Z}$ by $\mathbb{Z}$ is split (with splitting homomorphism μ mapping $\mathbb{Z}/n\mathbb{Z}$ isomorphically onto the second factor of the direct sum). On the other hand, the exact sequence of $\mathbb{Z}$ -modules $0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \xrightarrow{\pi} \mathbb{Z}/n\mathbb{Z} \rightarrow 0$ is not split since there is no nonzero homomorphism of $\mathbb{Z}/n\mathbb{Z}$ into $\mathbb{Z}$.
- (3) Neither D_8 nor Q_8 is a split extension of $\mathbb{Z}_2 \times \mathbb{Z}_2$ by $\mathbb{Z}_2$ because in neither group is there a subgroup complement to the center (Section 2.5 gives the subgroup structures of these groups).
- (4) The group D_8 is a split extension of $\mathbb{Z}_2$ by $\mathbb{Z}_4$, i.e., there is a split short exact sequence

$$1 \rightarrow \mathbb{Z}_4 \xrightarrow{\iota} D_8 \xrightarrow{\pi} \mathbb{Z}_2 \rightarrow 1,$$

namely,

$$1 \rightarrow \langle r \rangle \xrightarrow{\iota} D_8 \xrightarrow{\pi} \langle \bar{s} \rangle \rightarrow 1,$$

using our usual set of generators for D_8 . Here ι is the inclusion map and $\pi : r^a s^b \mapsto \bar{s}^b$ is the projection onto the quotient $D_8/\langle r \rangle \cong \mathbb{Z}_2$. The splitting homomorphism μ

maps $\langle \bar{s} \rangle$ isomorphically onto the complement $\langle s \rangle$ for $\langle r \rangle$ in D_8 . Equivalently, D_8 is the semidirect product of the normal subgroup $\langle r \rangle$ (isomorphic to Z_4) with $\langle s \rangle$ (isomorphic to Z_2).

On the other hand, while Q_8 is also an extension of Z_2 by Z_4 (for example, $\langle i \rangle \cong Z_4$ has quotient isomorphic to Z_2), Q_8 is *not* a split extension of Z_2 by Z_4 : no cyclic subgroup of Q_8 of order 4 has a complement in Q_8 .

Section 5.5 contains many more examples of split extensions of groups.

Proposition 25 shows that an extension B of C by A is a split extension if and only if there is a splitting homomorphism μ of the projection map $\varphi : B \rightarrow C$ from B to the quotient C . The next proposition shows in particular that for modules this is equivalent to the existence of a splitting homomorphism for ψ at the other end of the sequence.

Proposition 26. Let $0 \rightarrow A \xrightarrow{\psi} B \xrightarrow{\varphi} C \rightarrow 0$ be a short exact sequence of modules (respectively, $1 \rightarrow A \xrightarrow{\psi} B \xrightarrow{\varphi} C \rightarrow 1$ a short exact sequence of groups). Then $B = \psi(A) \oplus C'$ for some submodule C' of B with $\varphi(C') \cong C$ (respectively, $B = \psi(A) \times C'$ for some subgroup C' of B with $\varphi(C') \cong C$) if and only if there is a homomorphism $\lambda : B \rightarrow A$ such that $\lambda \circ \psi$ is the identity map on A .

Proof: This is similar to the proof of Proposition 25. If λ is given, define $C' = \ker \lambda \subseteq B$ and if C' is given define $\lambda : B = \psi(A) \oplus C' \rightarrow A$ by $\lambda((\psi(a), c')) = a$. Note that in this case $C' = \ker \lambda$ is *normal* in B , so that C' is a *normal* complement to $\psi(A)$ in B , which in turn implies that B is the *direct sum* of $\psi(A)$ and C' (cf. Theorem 9 of Section 5.4).

Proposition 26 shows that for general group extensions, the existence of a splitting homomorphism λ on the *left* end of the sequence is stronger than the condition that the extension splits: in this case the extension group is a *direct* product, and not just a *semidirect* product. The fact that these two notions are equivalent in the context of modules is again a reflection of the abelian nature of the underlying groups, where semidirect products are always direct products.

Modules and $\text{Hom}_R(D, _)$

Let R be a ring with 1 and suppose the R -module M is an extension of N by L , with

$$0 \longrightarrow L \xrightarrow{\psi} M \xrightarrow{\varphi} N \longrightarrow 0$$

the corresponding short exact sequence of R -modules. It is natural to ask whether properties for L and N imply related properties for the extension M . The first situation we shall consider is whether an R -module homomorphism from some fixed R -module D to either L or N implies there is also an R -module homomorphism from D to M .

The question of obtaining a homomorphism from D to M given a homomorphism from D to L is easily disposed of: if $f \in \text{Hom}_R(D, L)$ is an R -module homomorphism from D to L then the composite $f' = \psi \circ f$ is an R -module homomorphism from D to

M . The relation between these maps can be indicated pictorially by the commutative diagram

$$\begin{array}{ccc} & D & \\ f \downarrow & \searrow f' & \\ L & \xrightarrow{\psi} & M \end{array}$$

Put another way, composition with ψ induces a map

$$\begin{aligned} \psi' : \text{Hom}_R(D, L) &\longrightarrow \text{Hom}_R(D, M) \\ f &\longmapsto f' = \psi \circ f. \end{aligned}$$

Recall that, by Proposition 2, $\text{Hom}_R(D, L)$ and $\text{Hom}_R(D, M)$ are abelian groups.

Proposition 27. Let D, L and M be R -modules and let $\psi : L \rightarrow M$ be an R -module homomorphism. Then the map

$$\begin{aligned} \psi' : \text{Hom}_R(D, L) &\longrightarrow \text{Hom}_R(D, M) \\ f &\longmapsto f' = \psi \circ f \end{aligned}$$

is a homomorphism of abelian groups. If ψ is injective, then ψ' is also injective, i.e.,

$$\text{if } 0 \longrightarrow L \xrightarrow{\psi} M \text{ is exact,}$$

$$\text{then } 0 \longrightarrow \text{Hom}_R(D, L) \xrightarrow{\psi'} \text{Hom}_R(D, M) \text{ is also exact.}$$

Proof: The fact that ψ' is a homomorphism is immediate. If ψ is injective, then distinct homomorphisms f and g from D into L give distinct homomorphisms $\psi \circ f$ and $\psi \circ g$ from D into M , which is to say that ψ' is also injective.

While obtaining homomorphisms into M from homomorphisms into the submodule L is straightforward, the situation for homomorphisms into the quotient N is much less evident. More precisely, given an R -module homomorphism $f : D \rightarrow N$ the question is whether there exists an R -module homomorphism $F : D \rightarrow M$ that *extends* or *lifts* f to M , i.e., that makes the following diagram commute:

$$\begin{array}{ccc} & D & \\ F \swarrow & \downarrow f & \\ M & \xrightarrow{\varphi} & N \end{array}$$

As before, composition with the homomorphism φ induces a homomorphism of abelian groups

$$\begin{aligned} \varphi' : \text{Hom}_R(D, M) &\longrightarrow \text{Hom}_R(D, N) \\ F &\longmapsto F' = \varphi \circ F. \end{aligned}$$

In terms of φ' , the homomorphism f to N lifts to a homomorphism to M if and only if f is in the image of φ' (namely, f is the image of the lift F).